

BIOLOGY

TENTH EDITION

Global Edition

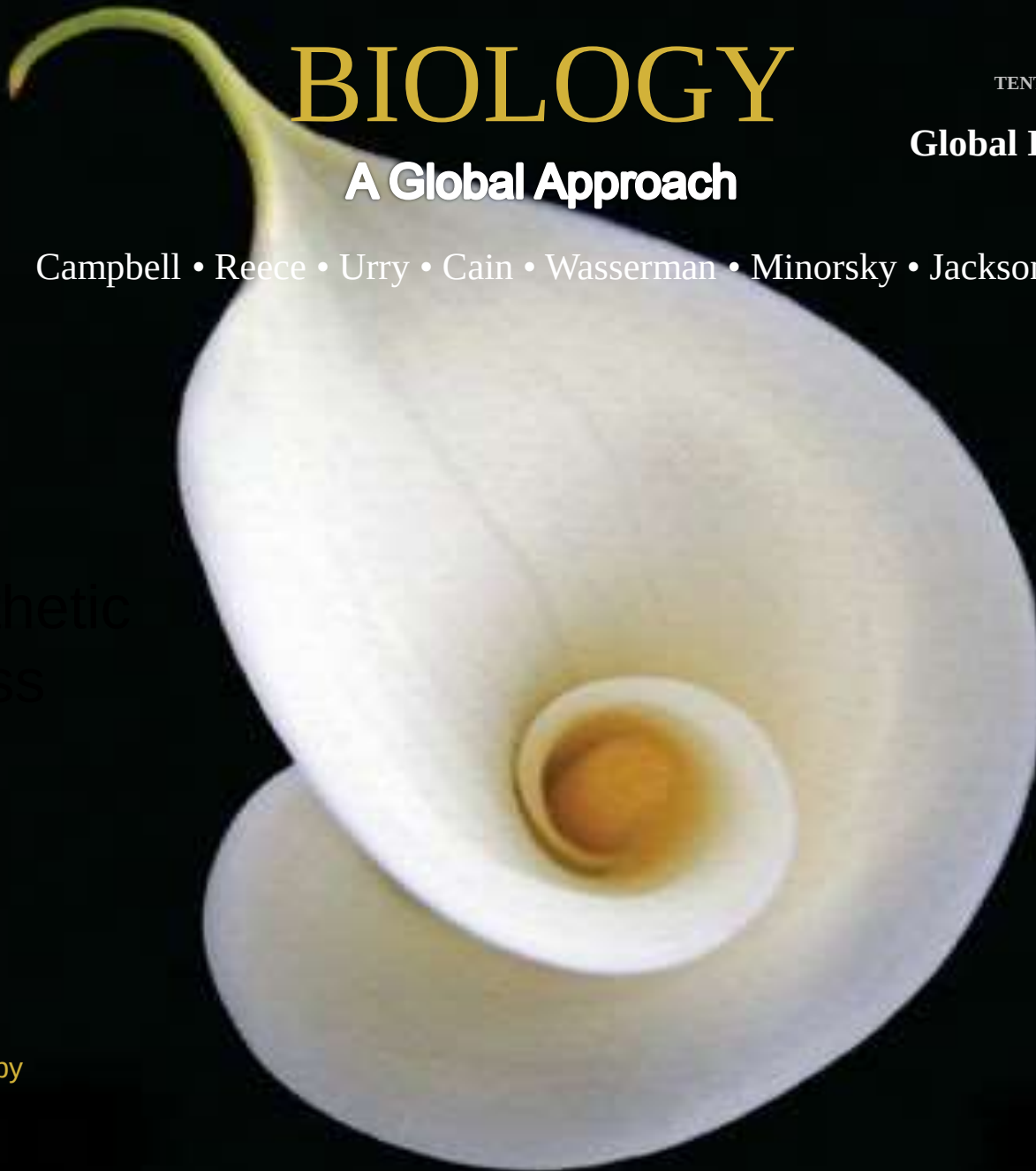
A Global Approach

Campbell • Reece • Urry • Cain • Wasserman • Minorsky • Jackson

11

Photosynthetic Process

Lecture Presentation by
Nicole Tunbridge and
Kathleen Fitzpatrick

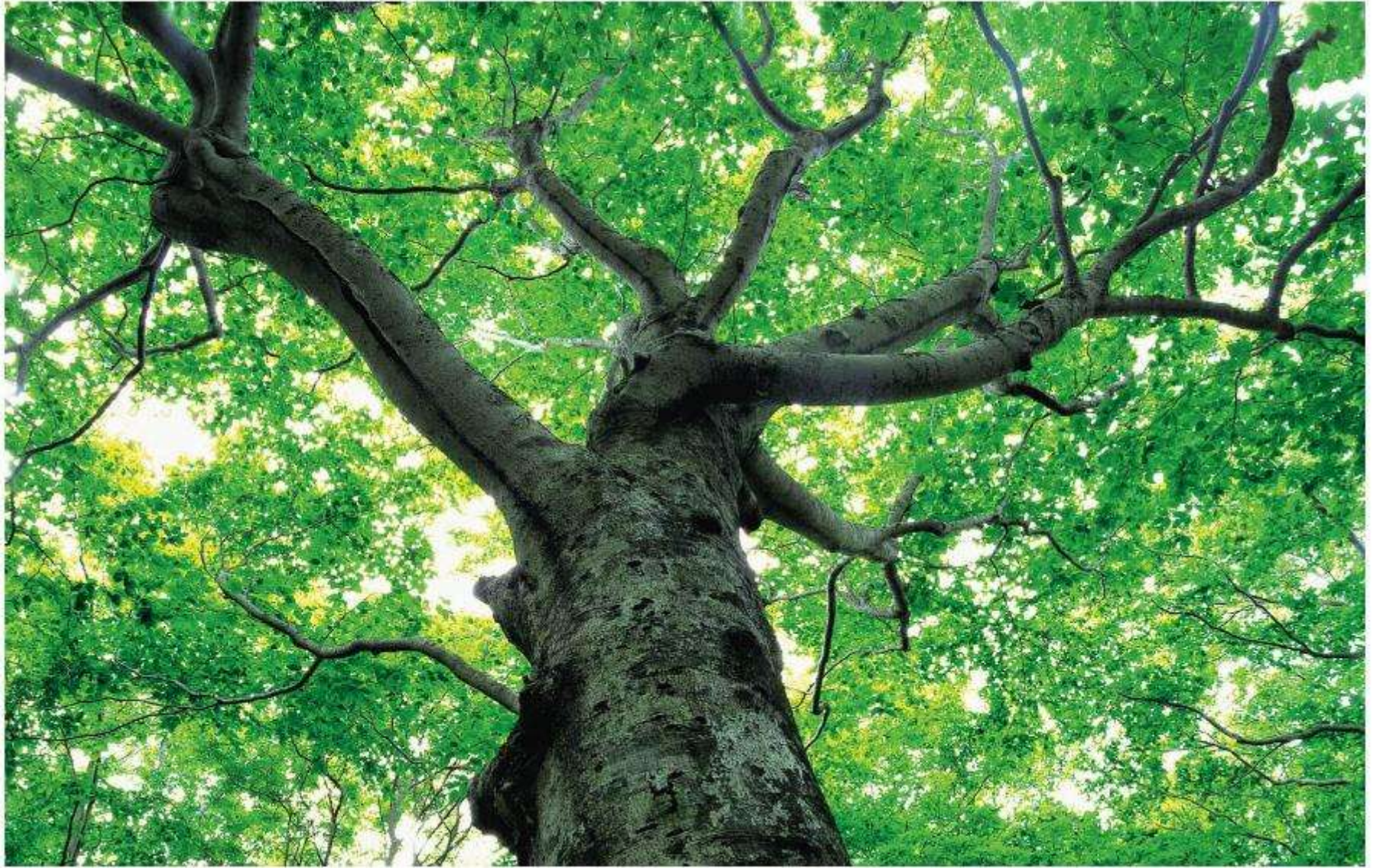


The Process That Feeds the Biosphere

- a) **Photosynthesis** is the process that converts solar energy into chemical energy
- b) Directly or indirectly, photosynthesis nourishes almost the entire living world

- a) **Autotrophs** sustain themselves without eating anything derived from other organisms
- b) Autotrophs are the producers of the biosphere, producing organic molecules from CO₂ and other inorganic molecules
- c) Almost all plants are photoautotrophs, using the energy of sunlight to make organic molecules

Figure 10.1





Other organisms also benefit from photosynthesis.

- a) Photosynthesis occurs in plants, algae, certain other unicellular eukaryotes, and some prokaryotes
- b) These organisms feed not only themselves but also most of the living world

Figure 10.2



(a) Plants

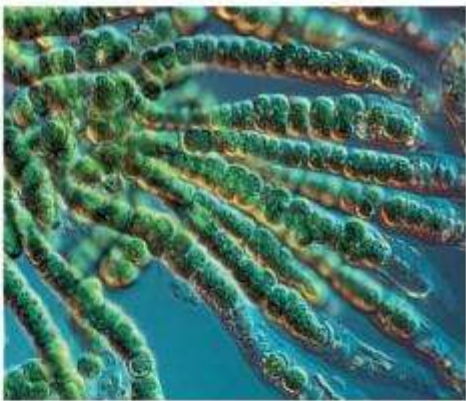


(b) Multicellular alga

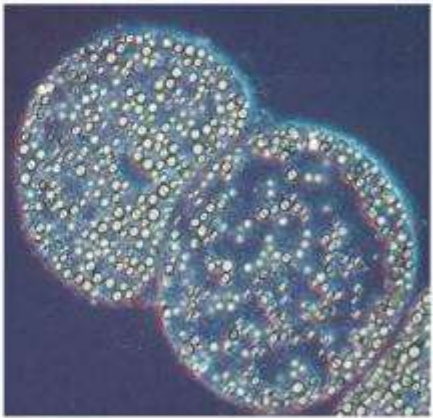


10 μm

(c) Unicellular eukaryotes



(d) Cyanobacteria 40 μm



1 μm

(e) Purple sulfur bacteria

- a) **Heterotrophs** obtain their organic material from other organisms
- b) Heterotrophs are the consumers of the biosphere
- c) Almost all heterotrophs, including humans, depend on photoautotrophs for food and O₂

- a) Earth's supply of fossil fuels was formed from the remains of organisms that died hundreds of millions of years ago
- b) In a sense, fossil fuels represent stores of solar energy from the distant past

Figure 10.3



Concept 10.1: Photosynthesis converts light energy to the chemical energy of food

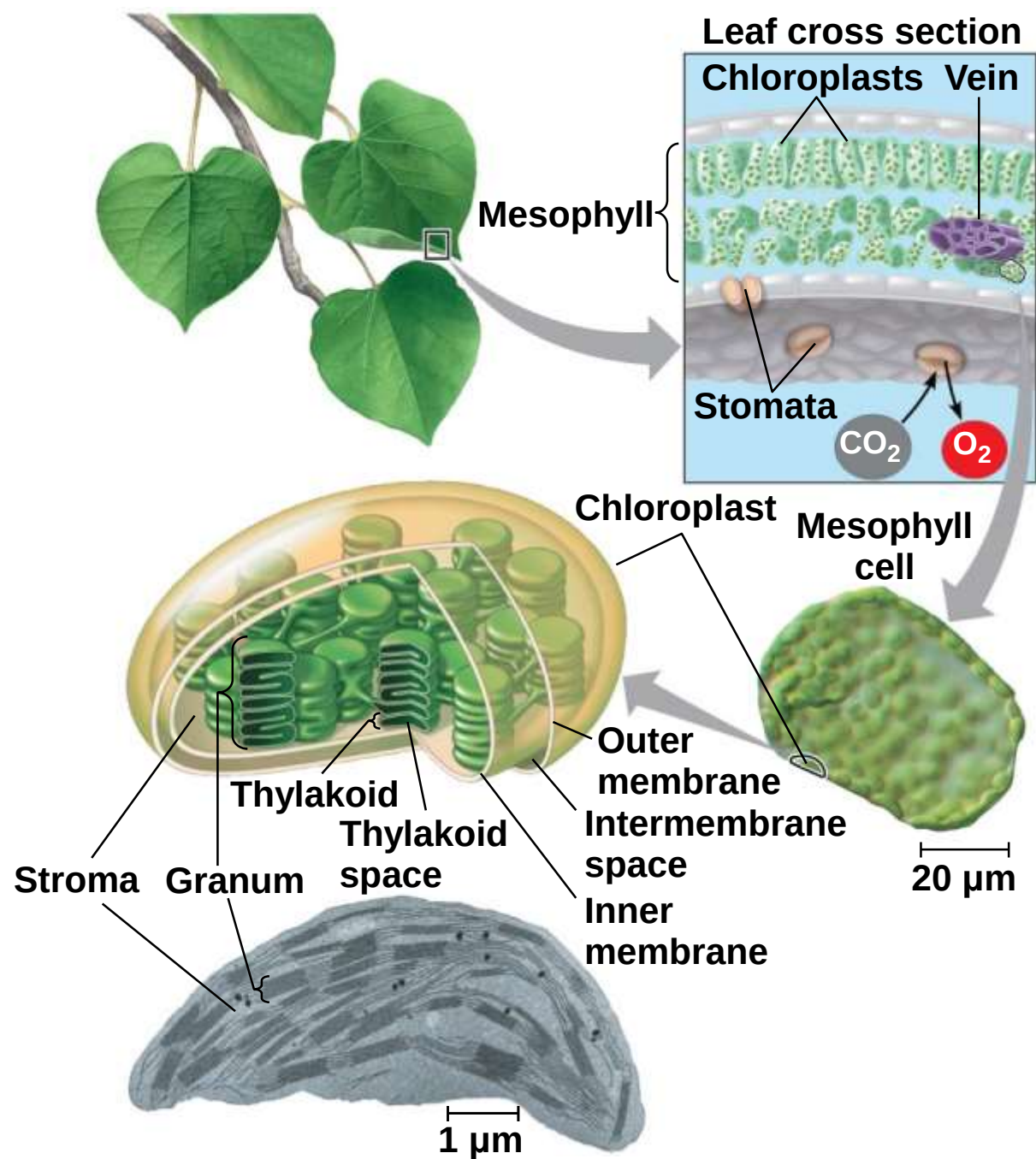
- a) Chloroplasts are structurally similar to and likely evolved from photosynthetic bacteria
- b) The structural organization of these organelles allows for the chemical reactions of photosynthesis

Chloroplasts: The Sites of Photosynthesis in Plants

- a) Leaves are the major locations of photosynthesis
- b) Chloroplasts are found mainly in cells of the mesophyll, the interior tissue of the leaf
- c) Each mesophyll cell contains 30–40 chloroplasts
- d) CO₂ enters and O₂ exits the leaf through microscopic pores called **stomata**

- a) A chloroplast has an envelope of two membranes surrounding a dense fluid called the **stroma**
- b) **Thylakoids** are connected sacs in the chloroplast which compose a third membrane system
- c) Thylakoids may be stacked in columns called grana
- d) **Chlorophyll**, the pigment which gives leaves their green colour, resides in the thylakoid membranes

Figure 10.4



Tracking Atoms Through Photosynthesis: *Scientific Inquiry*

a) Photosynthesis is a complex series of reactions that can be summarized as the following equation:

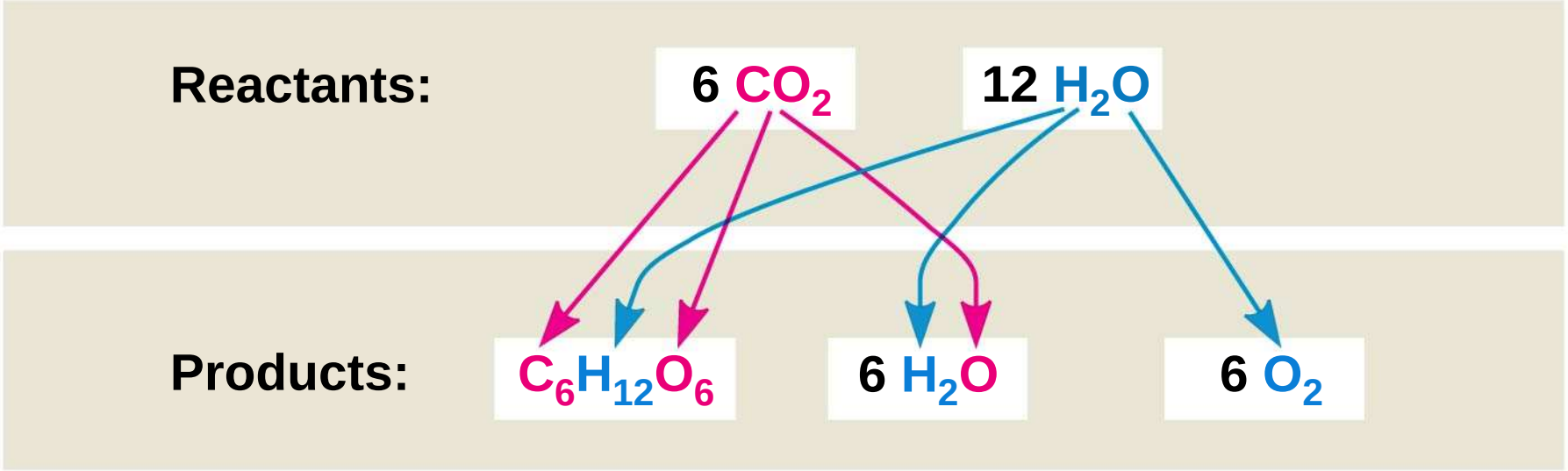


a) The overall chemical change during photosynthesis is the reverse of the one that occurs during cellular respiration

The Splitting of Water

- a) Chloroplasts split H_2O into hydrogen and oxygen, incorporating the electrons of hydrogen into sugar molecules and releasing oxygen as a by-product

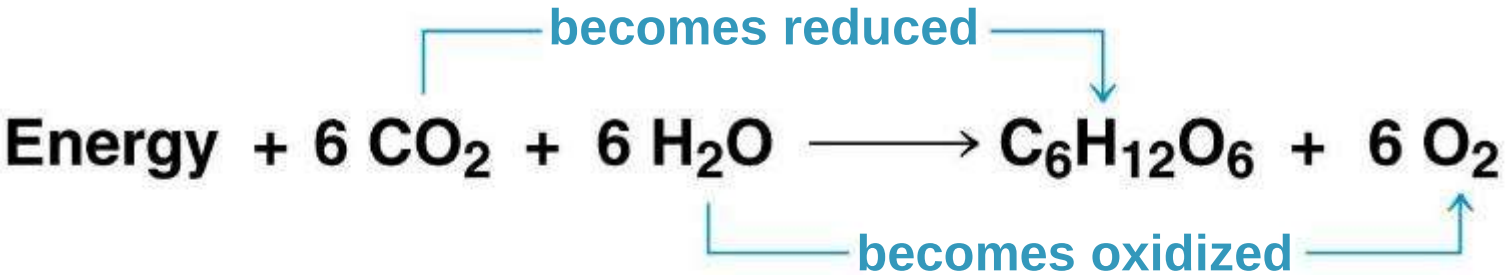
Figure 10.5



Photosynthesis as a Redox Process

- a) Photosynthesis reverses the direction of electron flow compared to respiration
- b) Photosynthesis is a redox process in which H_2O is oxidized and CO_2 is reduced
- c) Photosynthesis is an endergonic process; the energy boost is provided by light

Figure 10.UN01



The Two Stages of Photosynthesis: *A Preview*

- a) Photosynthesis consists of the **light reactions** (the *photo part*) and **Calvin cycle** (the *synthesis part*)
- b) The light reactions (in the thylakoids)
 - a) Split H_2O
 - b) Release O_2
 - c) Reduce the electron acceptor **NADP⁺** to NADPH
 - d) Generate ATP from ADP by **photophosphorylation**

- a) The Calvin cycle (in the stroma) forms sugar from CO₂, using ATP and NADPH
- b) The Calvin cycle begins with **carbon fixation**, incorporating CO₂ into organic molecules

Figure 10.6-1

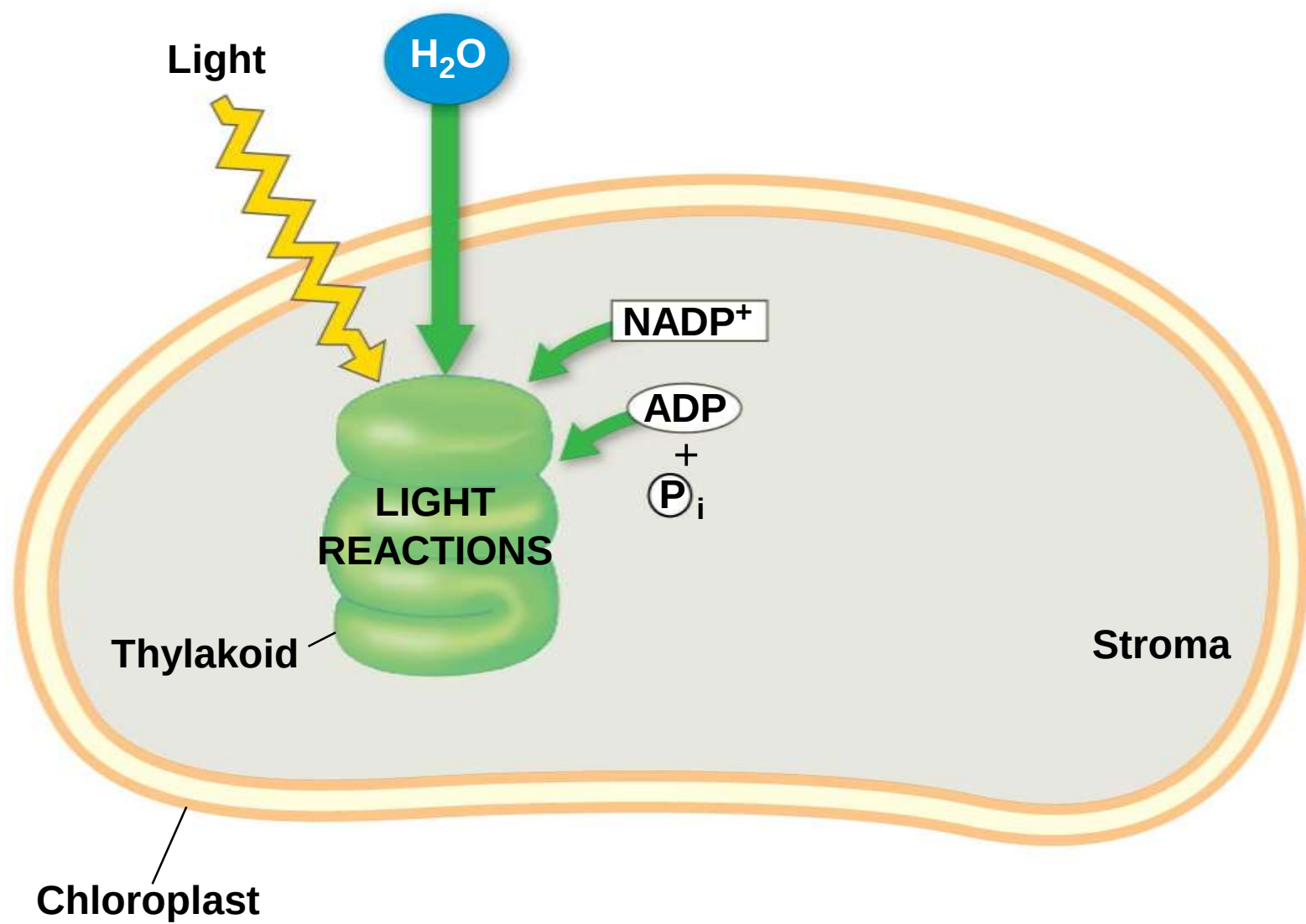


Figure 10.6-2

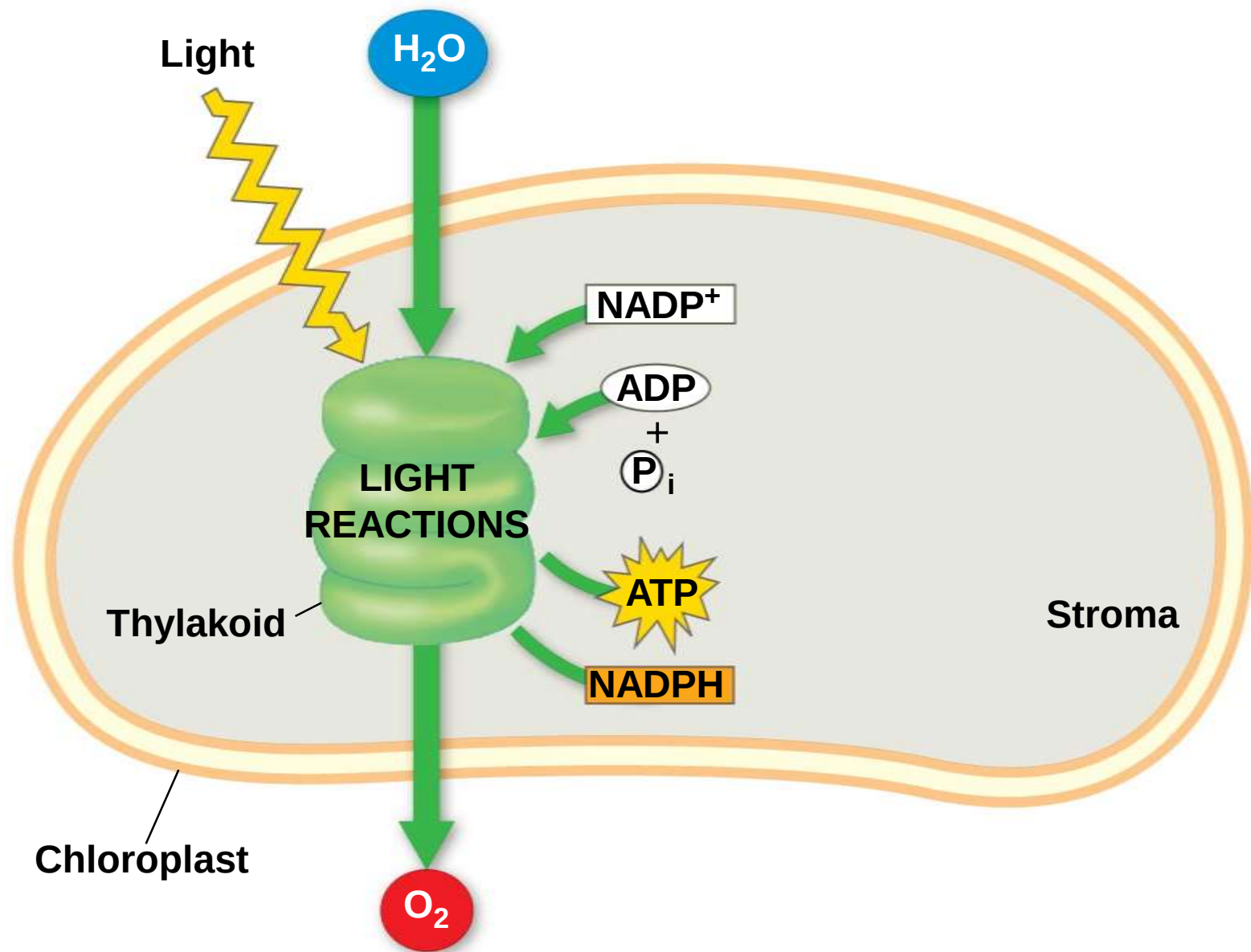


Figure 10.6-3

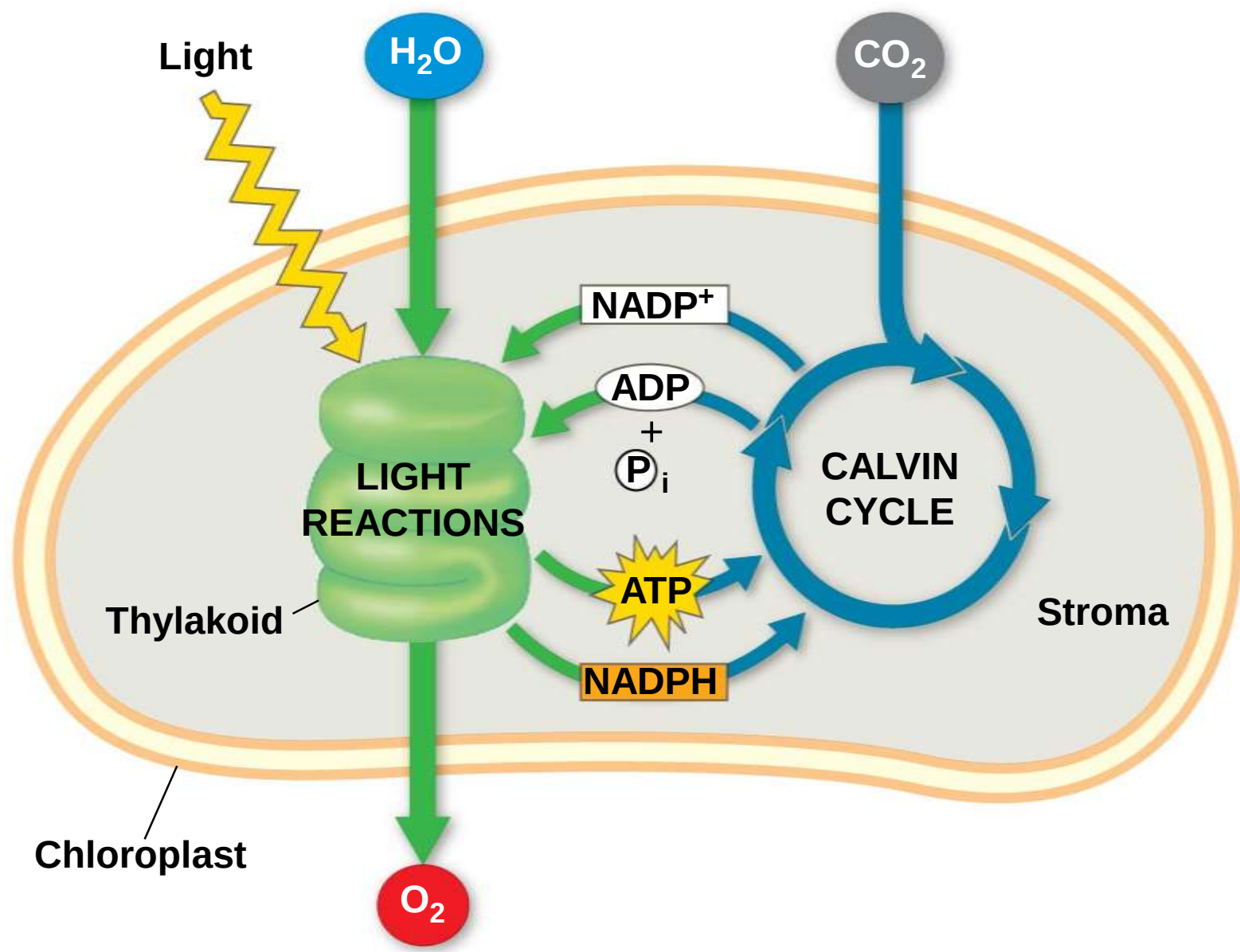
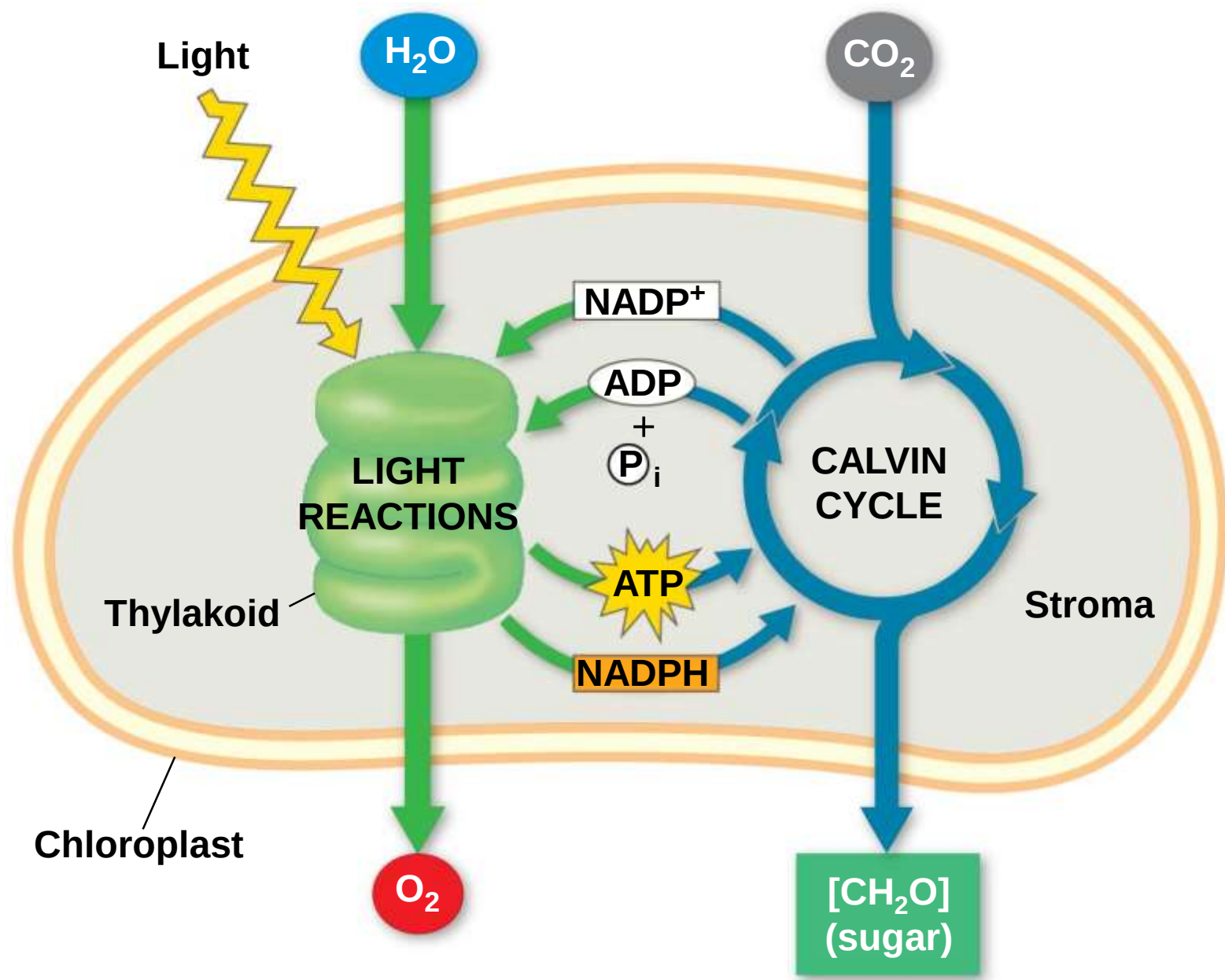


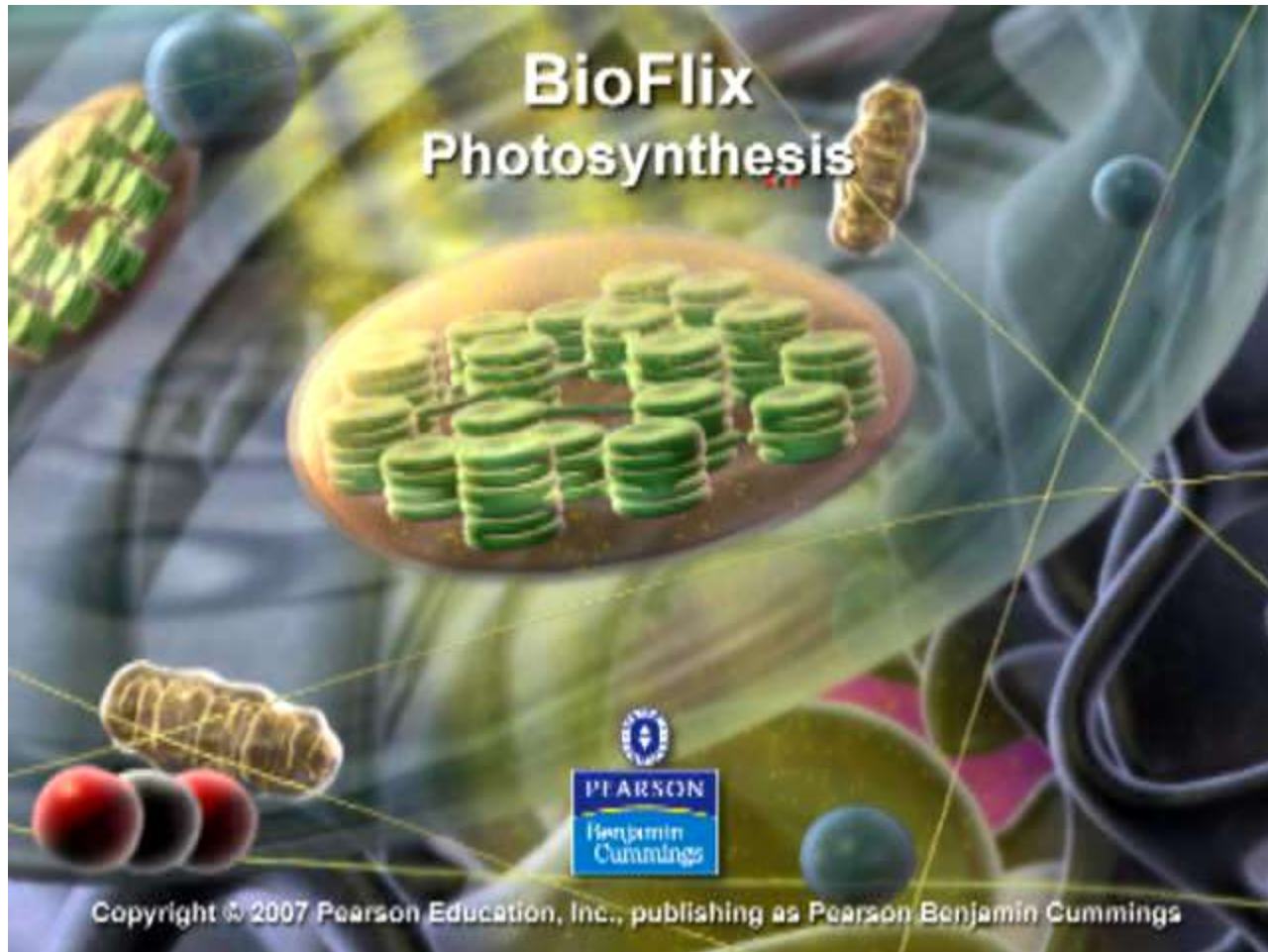
Figure 10.6-4



BioFlix: The Carbon Cycle



BioFlix: Photosynthesis



Concept 10.2: The light reactions convert solar energy to the chemical energy of ATP and NADPH

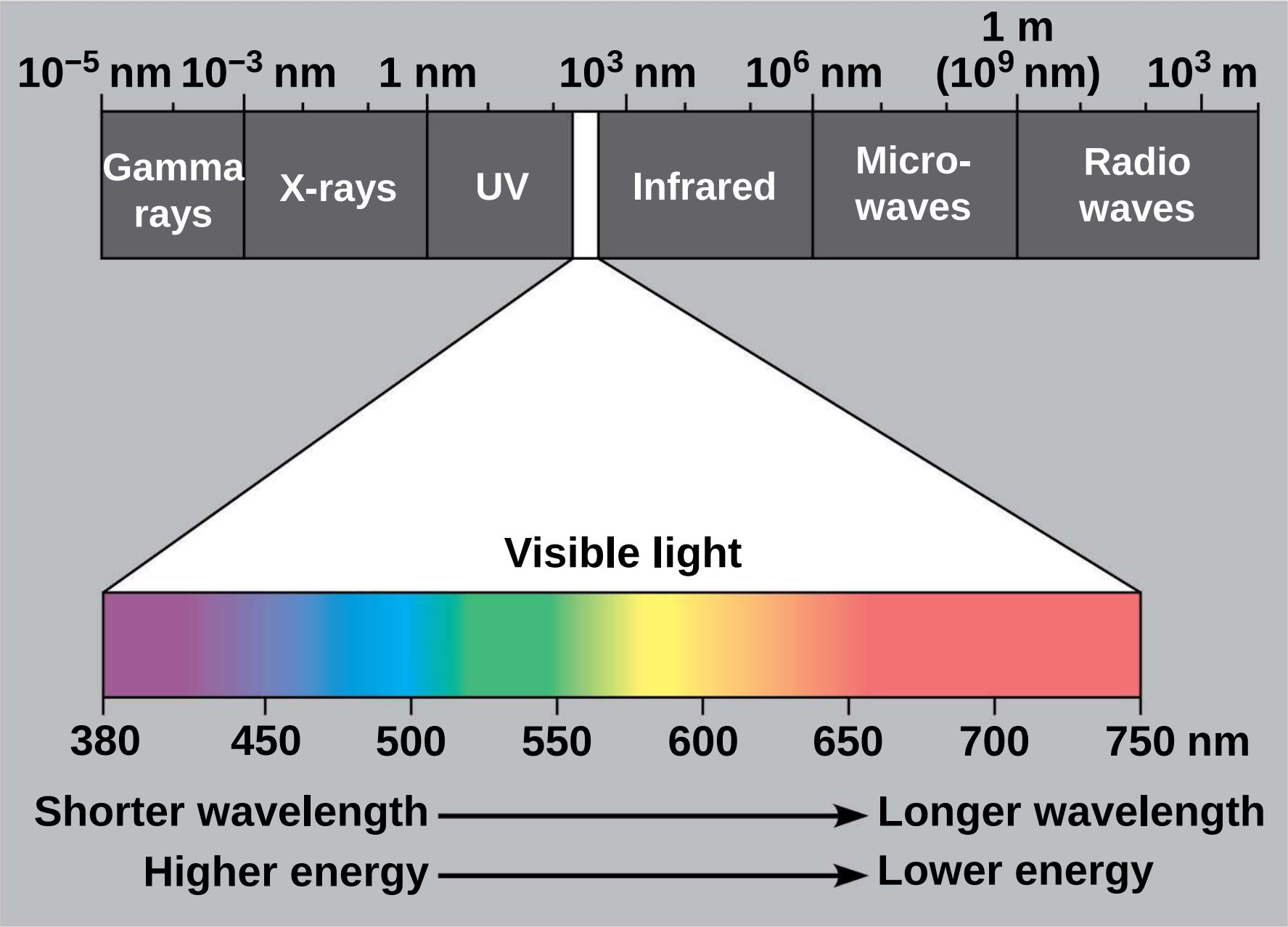
- a) Chloroplasts are solar-powered chemical factories
- b) Their thylakoids transform light energy into the chemical energy of ATP and NADPH

The Nature of Sunlight

- a) Light is a form of electromagnetic energy, also called electromagnetic radiation
- b) Like other electromagnetic energy, light travels in rhythmic waves
- c) **Wavelength** is the distance between crests of waves
- d) Wavelength determines the type of electromagnetic energy

- a) The **electromagnetic spectrum** is the entire range of electromagnetic energy, or radiation
- b) **Visible light** consists of wavelengths (including those that drive photosynthesis) that produce colors we can see
- c) Light also behaves as though it consists of discrete particles, called **photons**

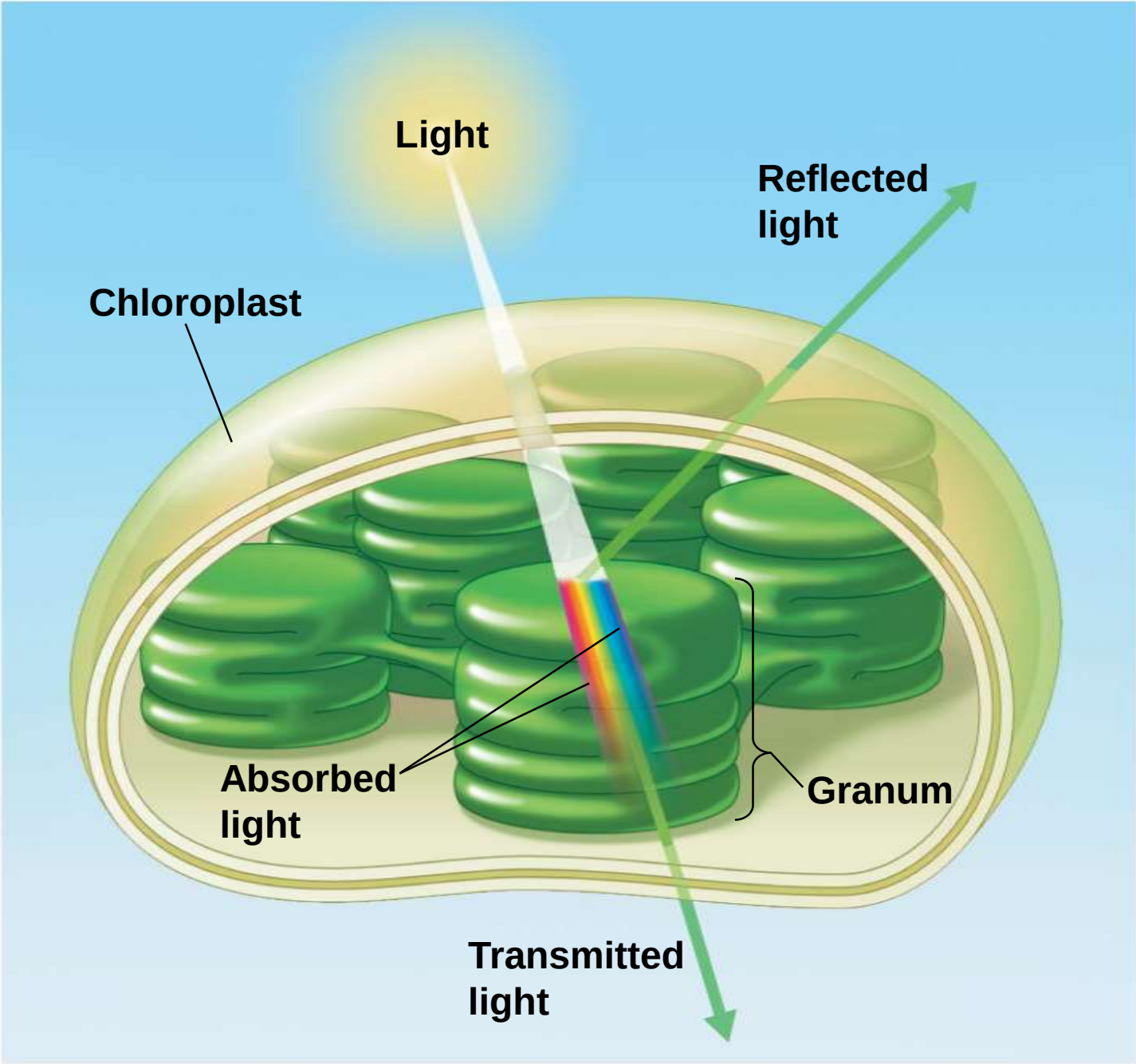
Figure 10.7



Photosynthetic Pigments: The Light Receptors

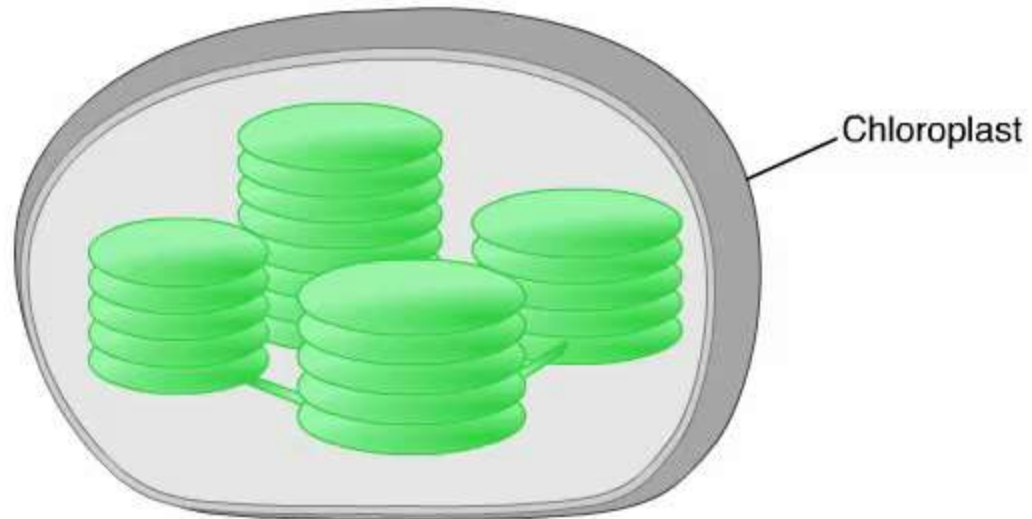
- a) Pigments are substances that absorb visible light
- b) Different pigments absorb different wavelengths
- c) Wavelengths that are not absorbed are reflected or transmitted
- d) Leaves appear green because chlorophyll reflects and transmits green light

Figure 10.8



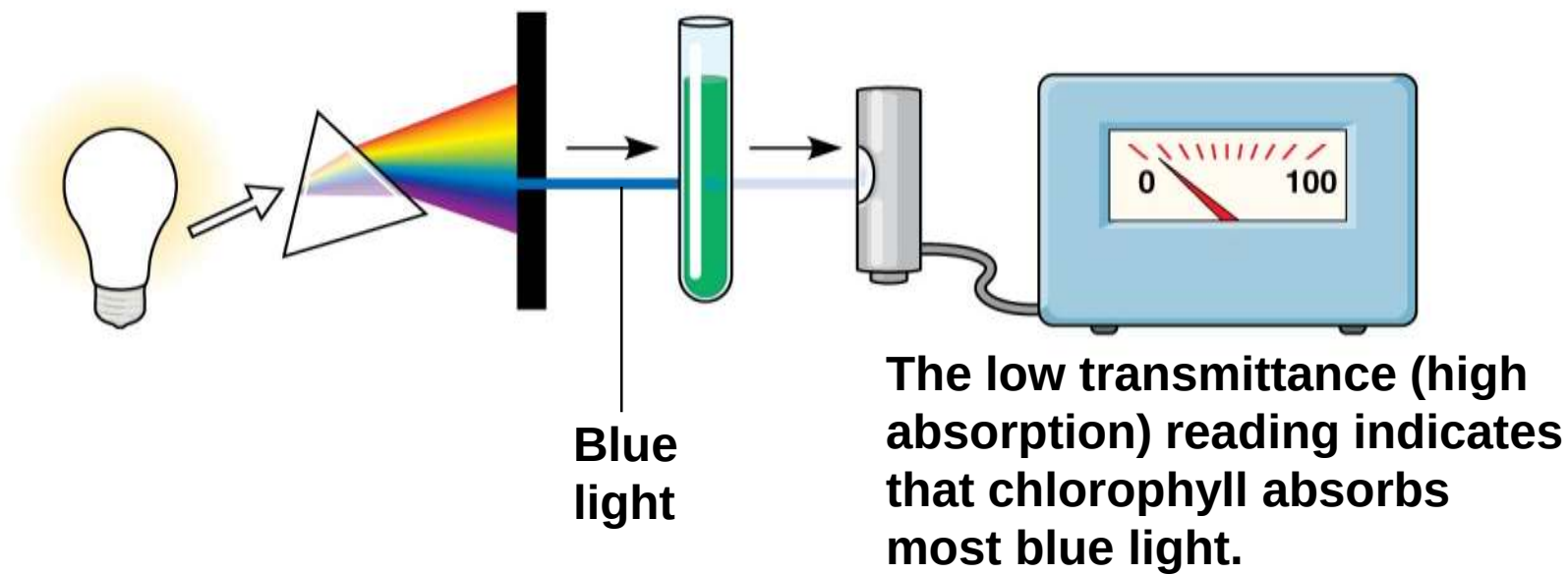
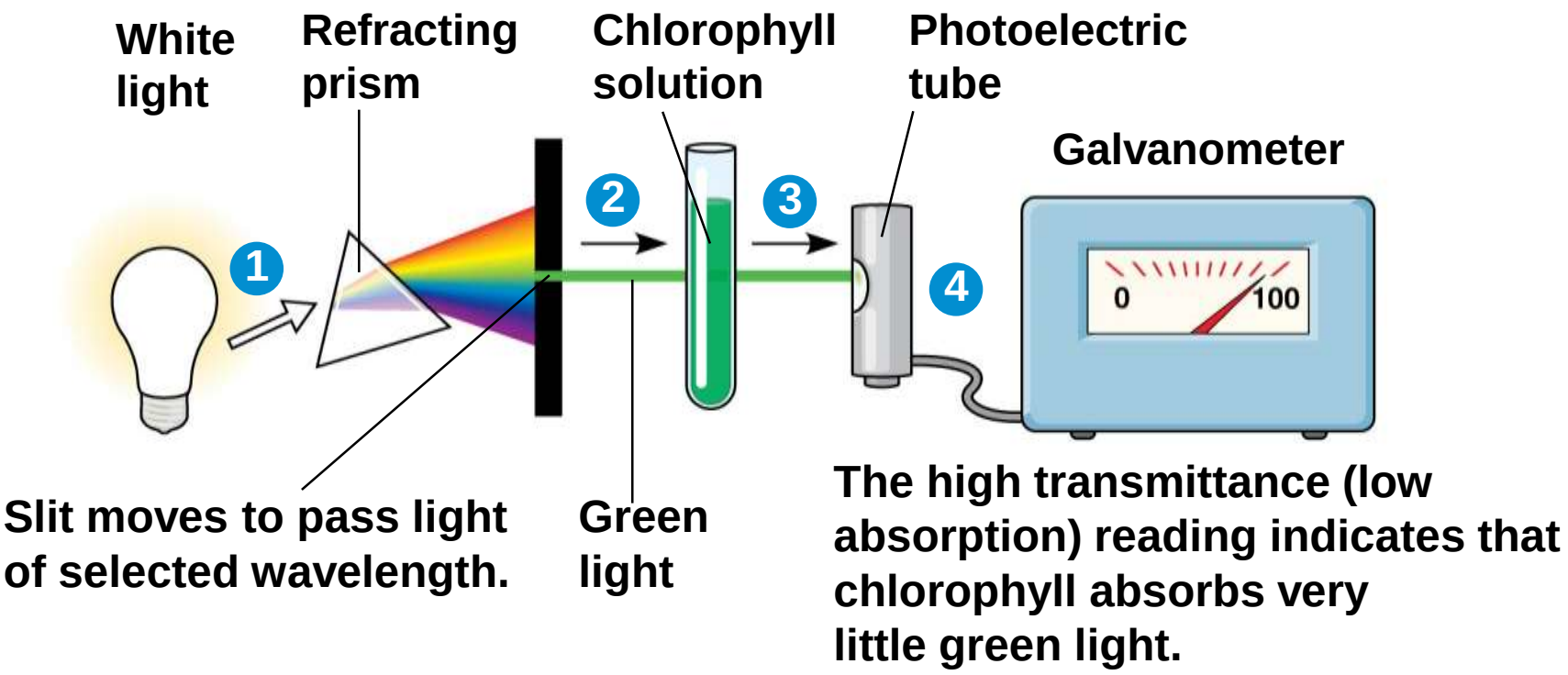
Animation: Light and Pigments

Light



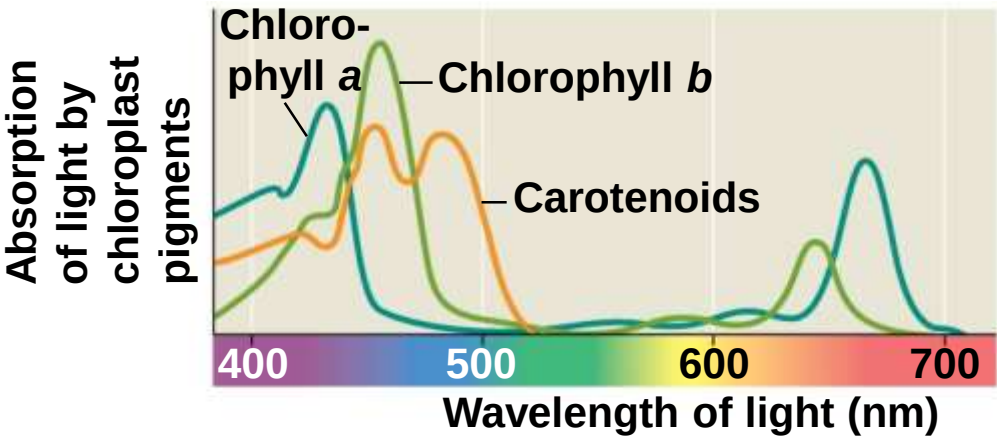
- a) A **spectrophotometer** measures a pigment's ability to absorb various wavelengths
- b) This machine sends light through pigments and measures the fraction of light transmitted at each wavelength

Figure 10.9

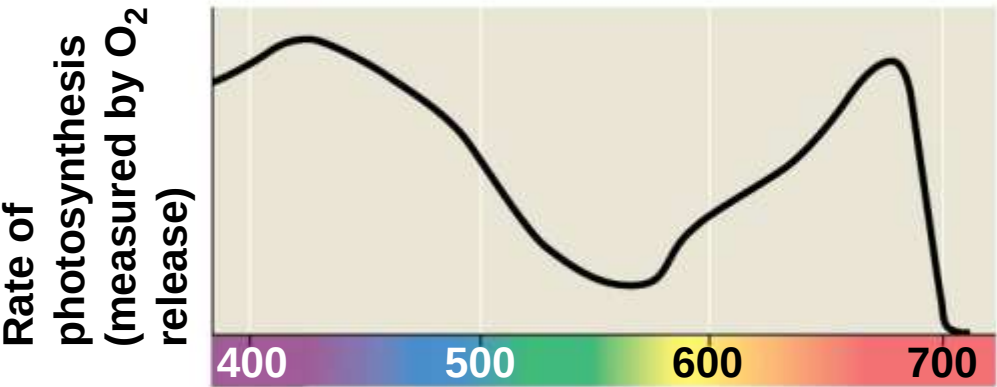


- a) An **absorption spectrum** is a graph plotting a pigment's light absorption versus wavelength
- b) The absorption spectrum of **chlorophyll a** suggests that violet-blue and red light work best for photosynthesis
- c) An **action spectrum** profiles the relative effectiveness of different wavelengths of radiation in driving a process

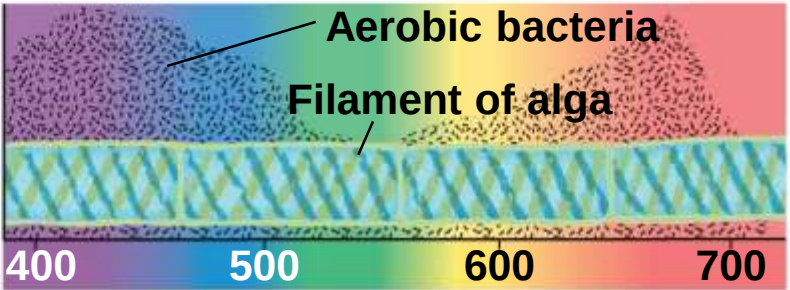
Figure 10.10



(a) Absorption spectra

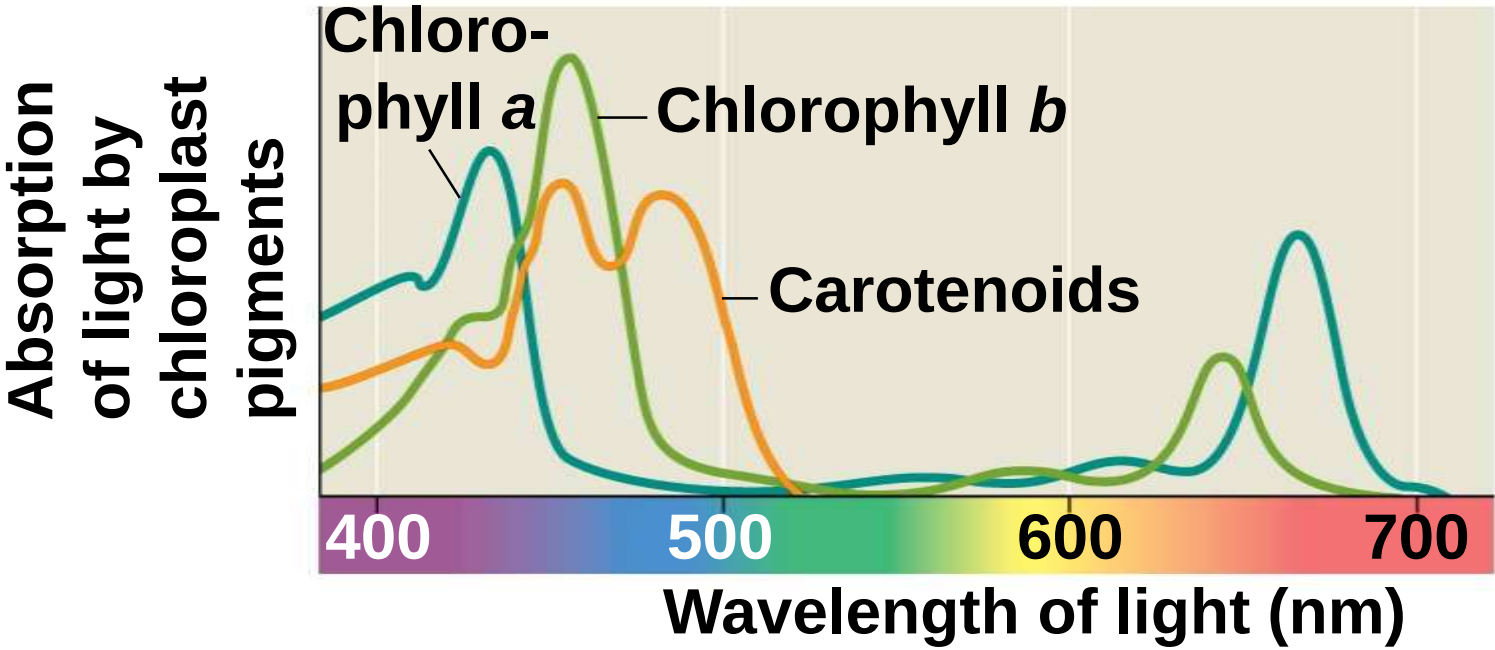


(b) Action spectrum



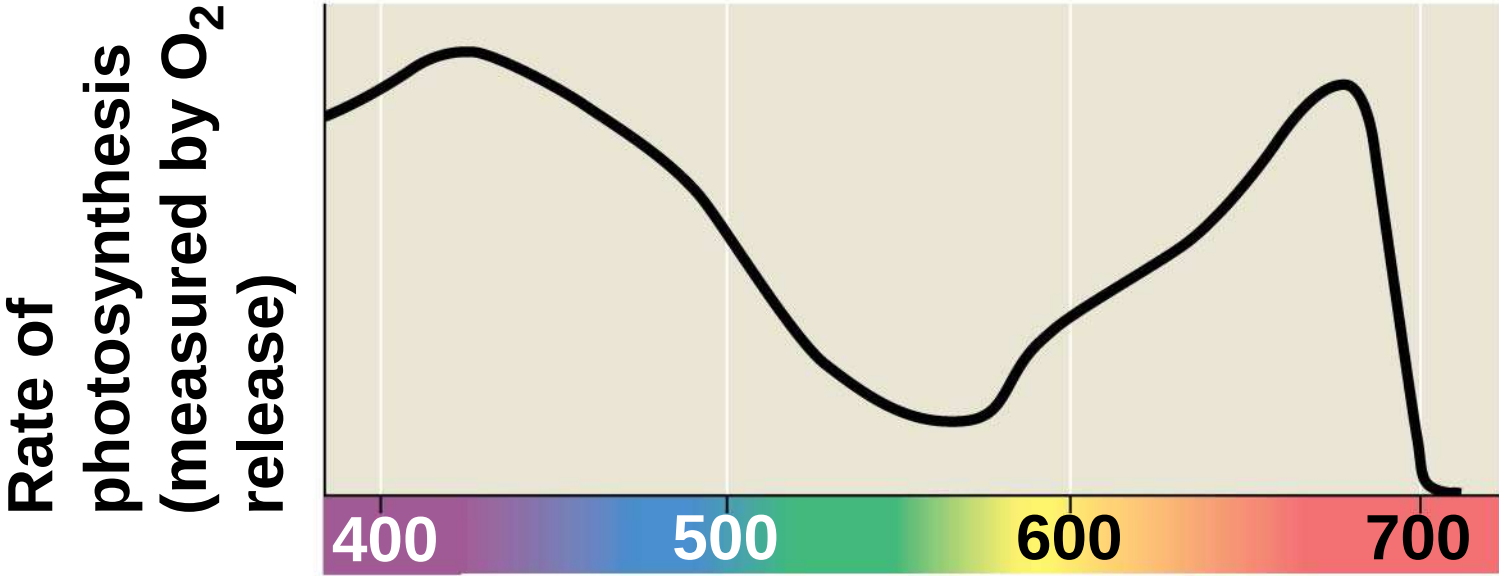
(c) Engelmann's experiment

Figure 10.10a



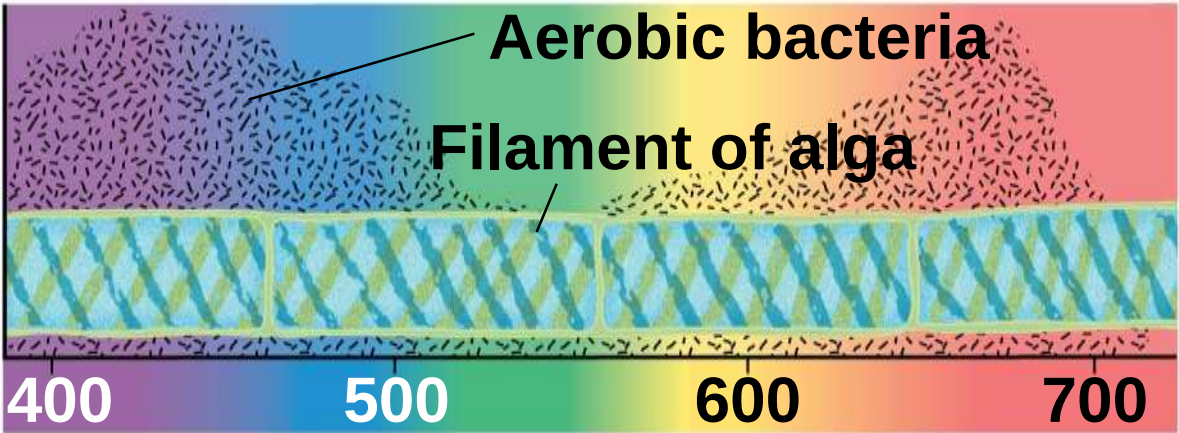
(a) Absorption spectra

Figure 10.10b



(b) Action spectrum

Figure 10.10c

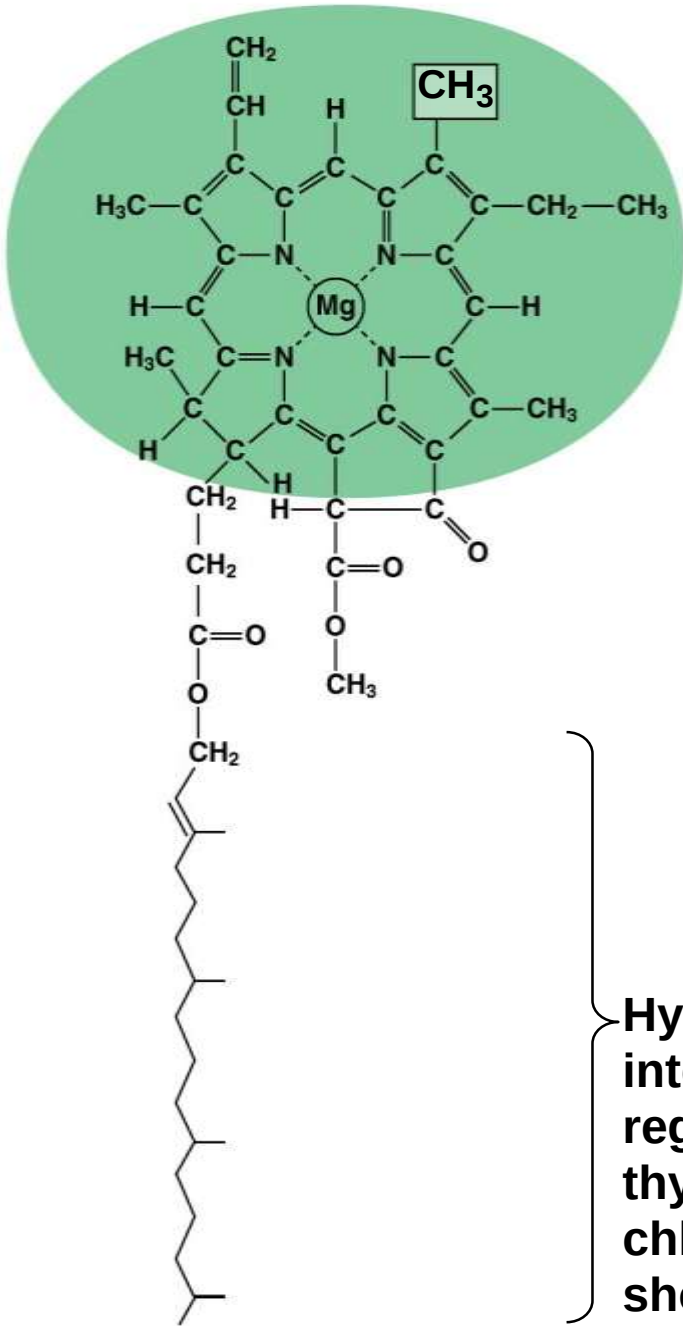


(c) Engelmann's experiment

- a) The action spectrum of photosynthesis was first demonstrated in 1883 by Theodor W. Engelmann
- b) In his experiment, he exposed different segments of a filamentous alga to different wavelengths
- c) Areas receiving wavelengths favorable to photosynthesis produced excess O₂
- d) He used the growth of aerobic bacteria clustered along the alga as a measure of O₂ production

- a) Chlorophyll *a* is the main photosynthetic pigment
- b) Accessory pigments, such as **chlorophyll *b***, broaden the spectrum used for photosynthesis
- c) The difference in the absorption spectrum between chlorophyll *a* and *b* is due to a slight structural difference between the pigment molecules
- d) Accessory pigments called **carotenoids** absorb excessive light that would damage chlorophyll

Figure 10.11

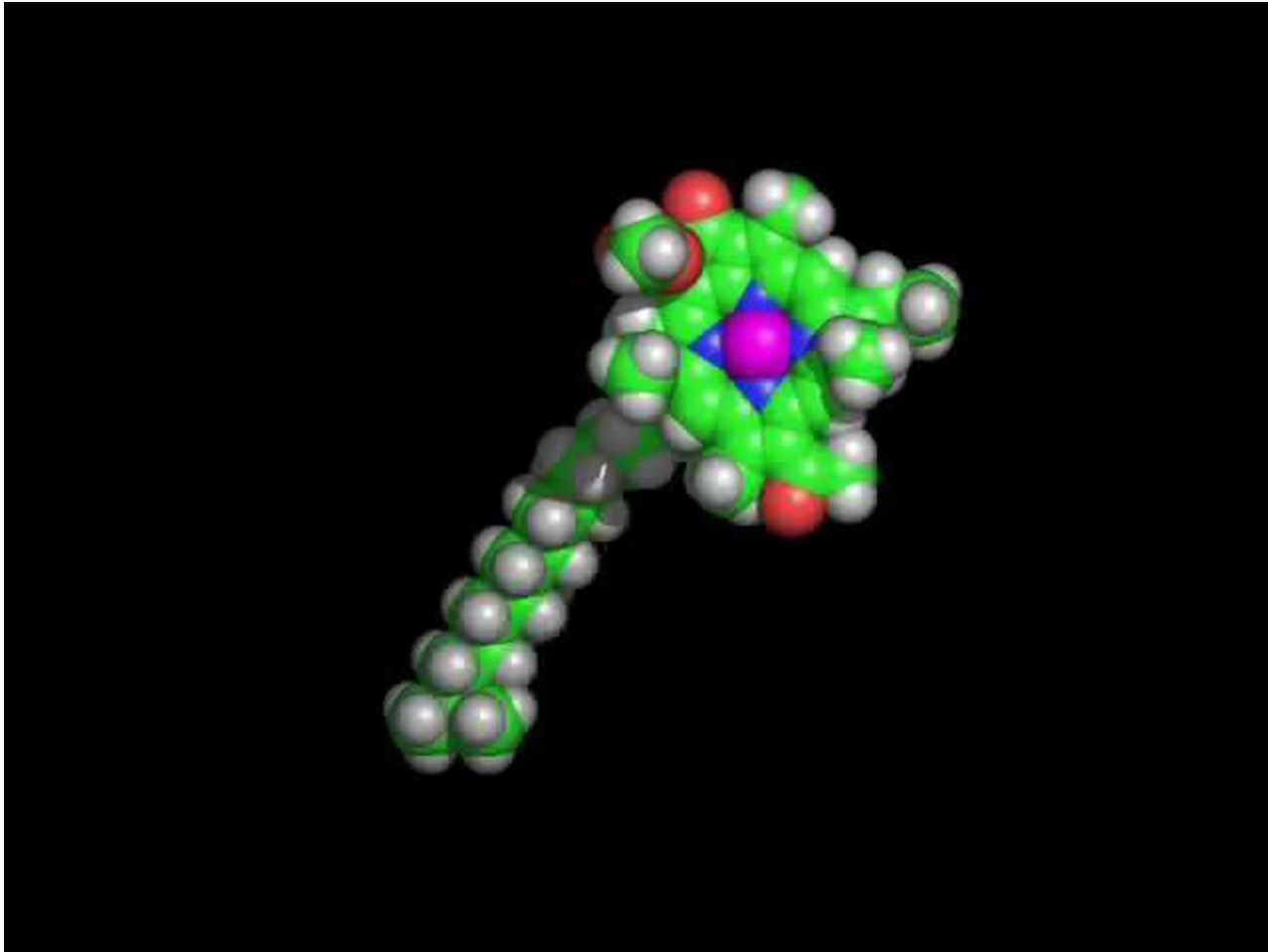


CH₃ in chlorophyll *a*
CHO in chlorophyll *b*

Porphyrin ring:
light-absorbing
“head” of molecule;
note magnesium
atom at center

Hydrocarbon tail:
interacts with hydrophobic
regions of proteins inside
thylakoid membranes of
chloroplasts; H atoms not
shown

Video: Space-Filling Model of Chlorophyll *a*

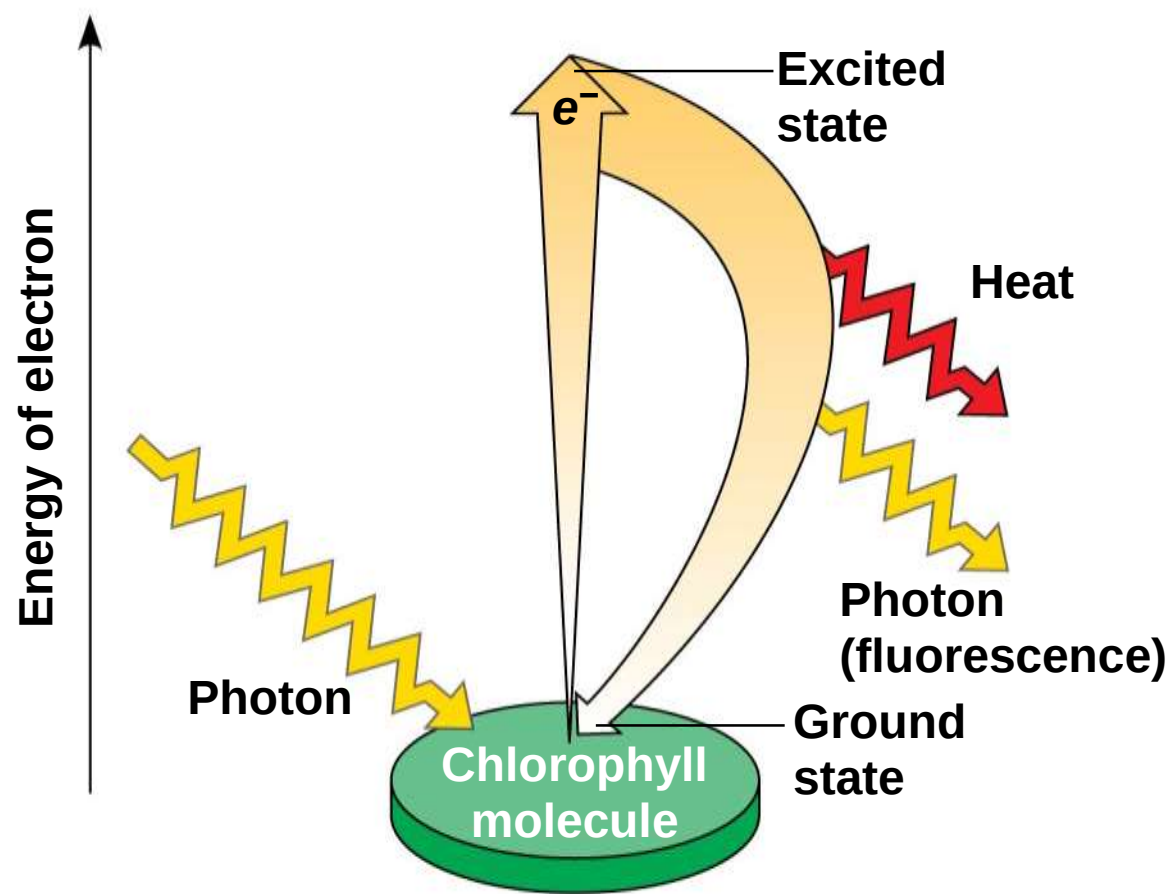


a) Accessory pigments called **carotenoids** function in photoprotection; they absorb excessive light that would damage chlorophyll

Excitation of Chlorophyll by Light

- a) When a pigment absorbs light, it goes from a ground state to an excited state, which is unstable
- b) When excited electrons fall back to the ground state, photons are given off, an afterglow called **fluorescence**
- c) If illuminated, an isolated solution of chlorophyll will fluoresce, giving off light and heat

Figure 10.12



(a) Excitation of isolated chlorophyll molecule

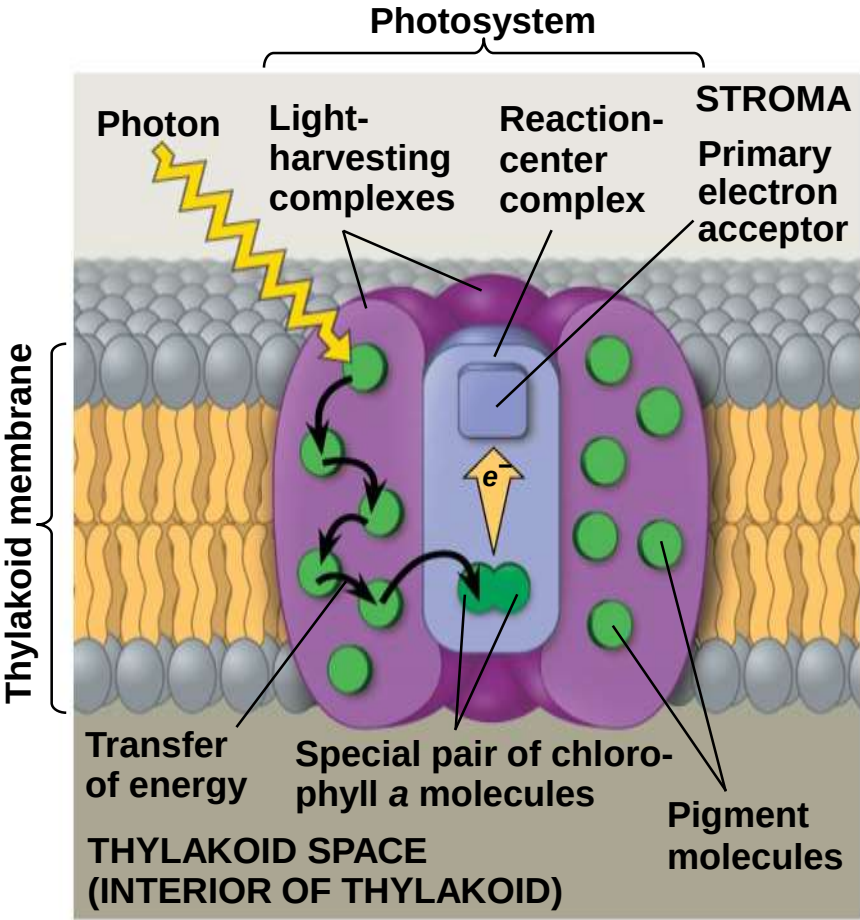


(b) Fluorescence

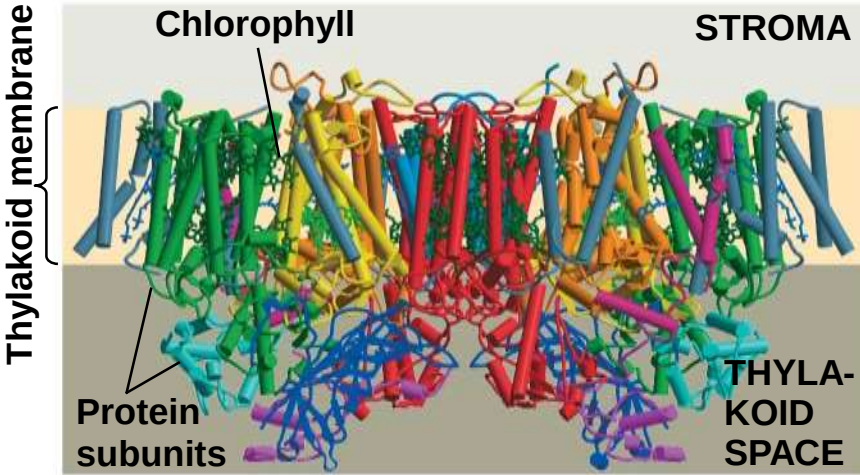
A Photosystem: A Reaction-Center Complex Associated with Light-Harvesting Complexes

- a) A **photosystem** consists of a **reaction-center complex** (a type of protein complex) surrounded by light-harvesting complexes
- b) The **light-harvesting complexes** (pigment molecules bound to proteins) transfer the energy of photons to the reaction center

Figure 10.13

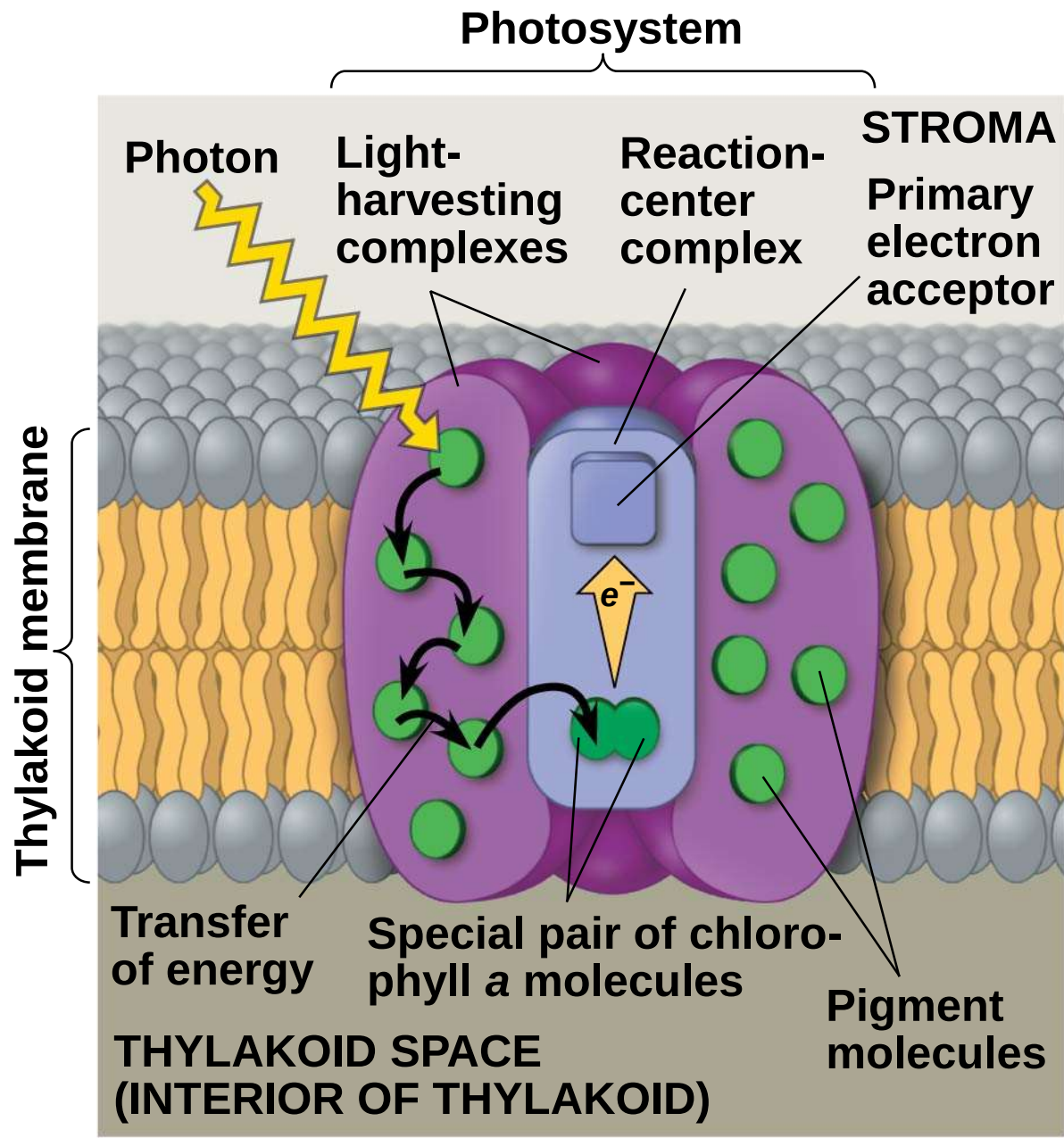


(a) How a photosystem harvests light



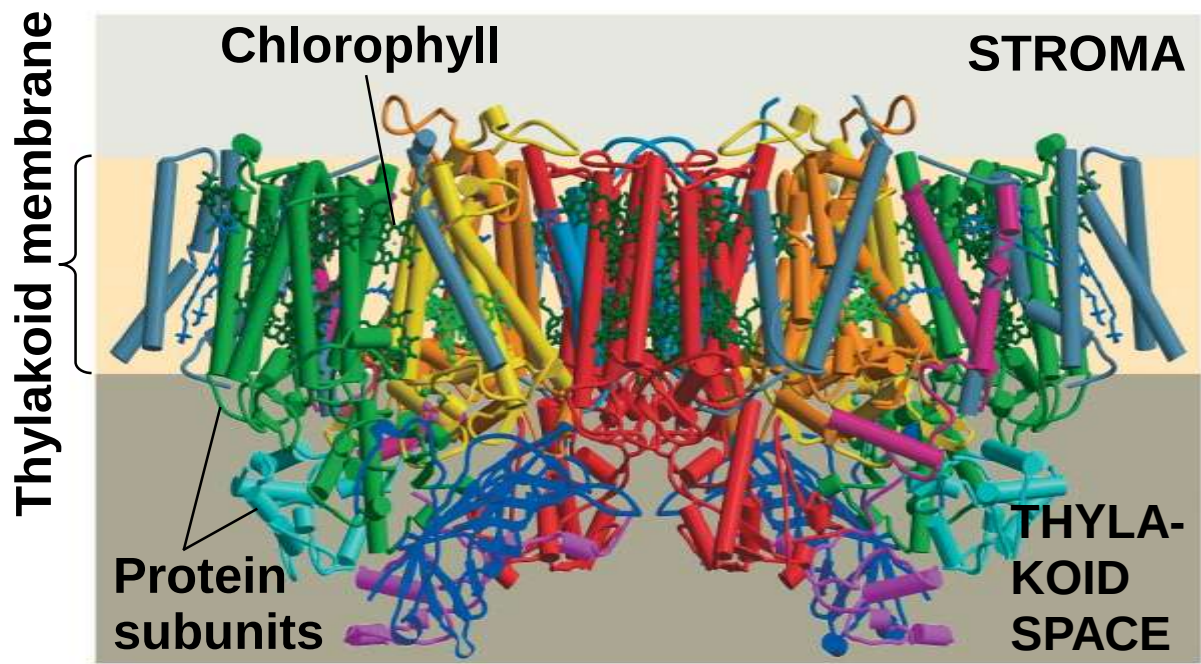
(b) Structure of a photosystem

Figure 10.13a



(a) How a photosystem harvests light

Figure 10.13b



(b) Structure of a photosystem

- a) A **primary electron acceptor** in the reaction center accepts excited electrons and is reduced as a result
- b) Solar-powered transfer of an electron from a chlorophyll *a* molecule to the primary electron acceptor is the first step of the light reactions

- a) There are two types of photosystems in the thylakoid membrane
- b) Photosystem II (PS II)** functions first (the numbers reflect order of discovery) and is best at absorbing a wavelength of 680 nm
- c) The reaction-center chlorophyll a of PS II is called P680

- a) **Photosystem I (PS I)** is best at absorbing a wavelength of 700 nm
- b) The reaction-center chlorophyll a of PS I is called P700

Linear Electron Flow

- a) During the light reactions, there are two possible routes for electron flow: cyclic and linear
- b) Linear electron flow**, the primary pathway, involves both photosystems and produces ATP and NADPH using light energy

a) There are 8 steps in linear electron flow:

1. A photon hits a pigment and its energy is passed among pigment molecules until it excites P680
2. An excited electron from P680 is transferred to the primary electron acceptor (we now call it P680⁺)

Figure 10.UN02

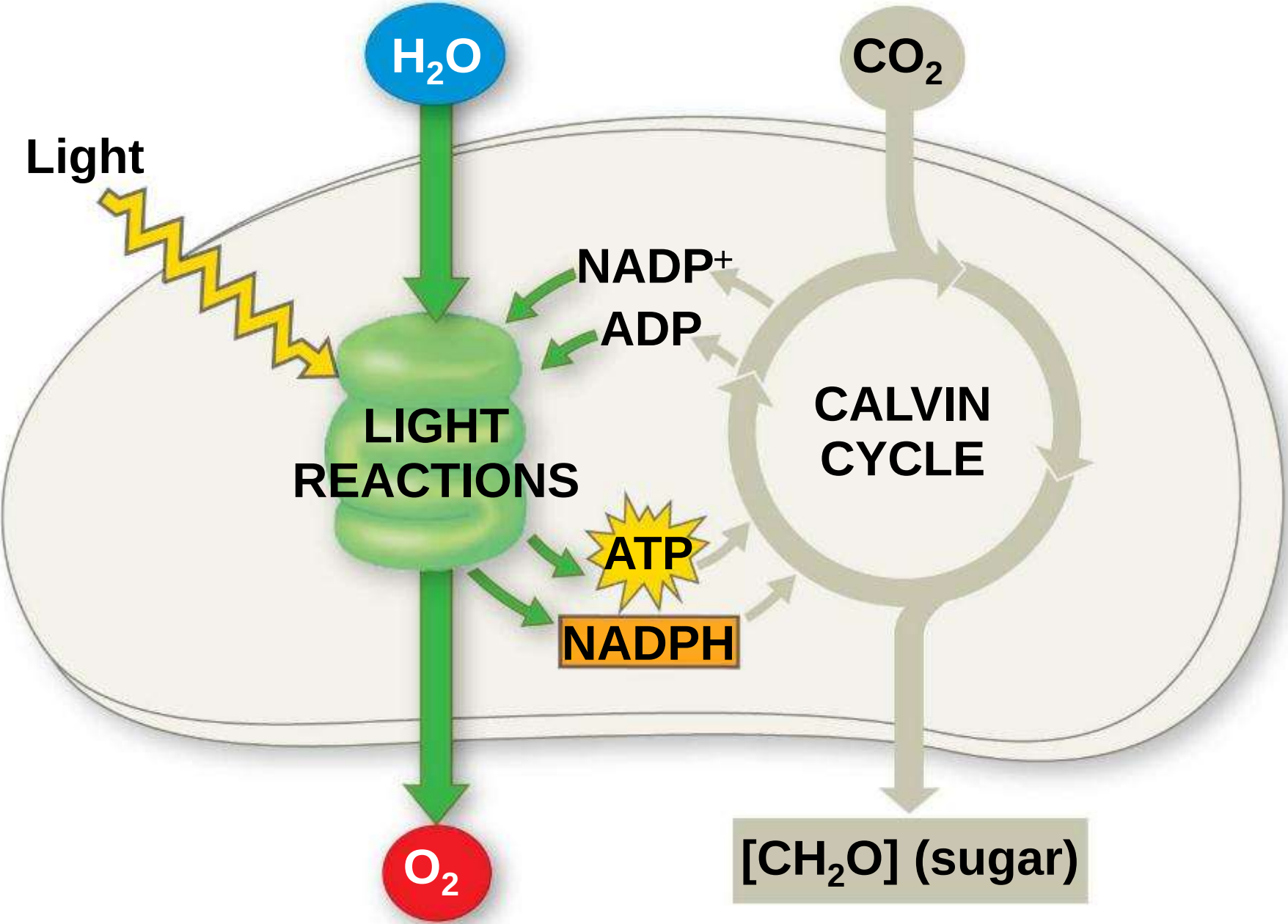
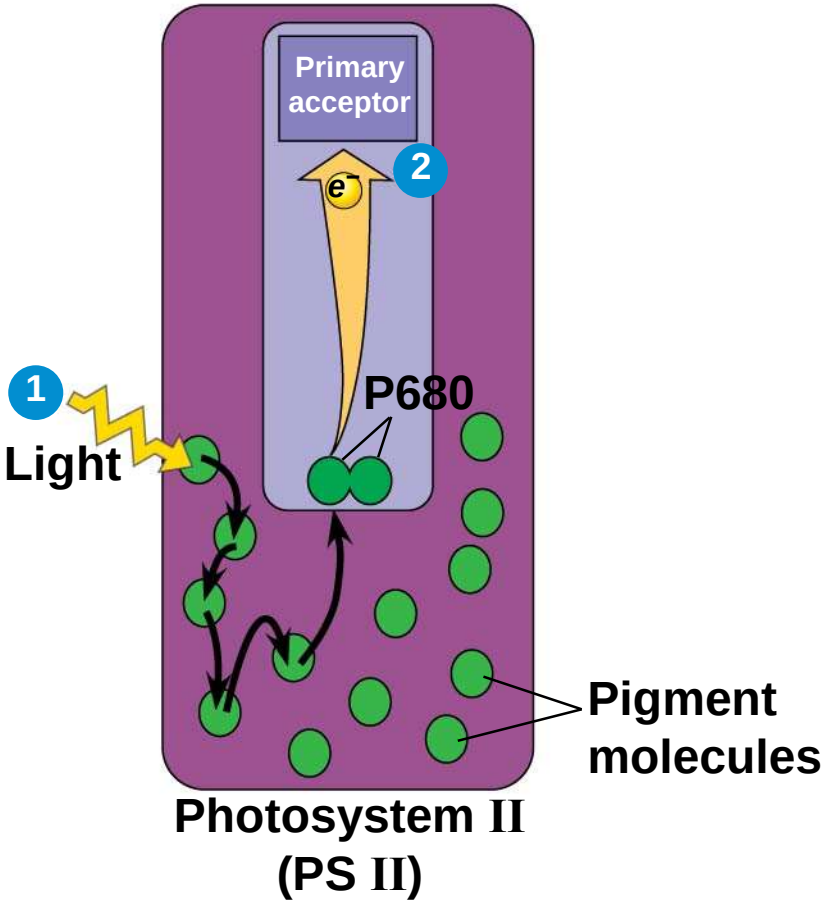


Figure 10.14-1

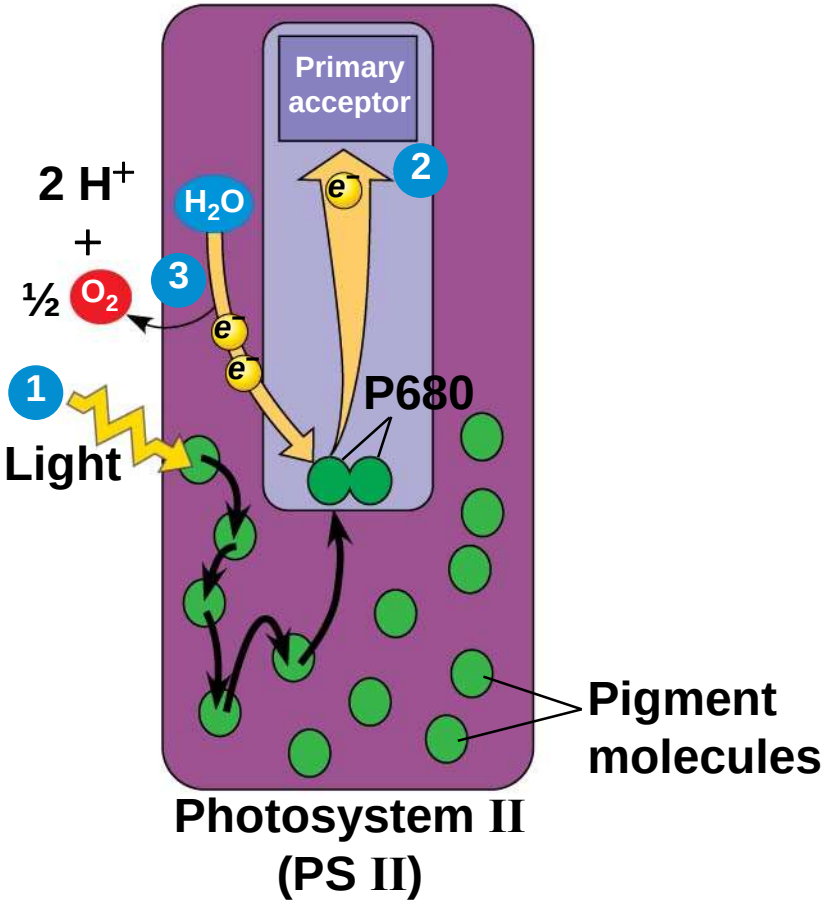


3. H_2O is split by enzymes, and the electrons are transferred from the hydrogen atoms to P680^+ , thus reducing it to P680

P680^+ is the strongest known biological oxidizing agent

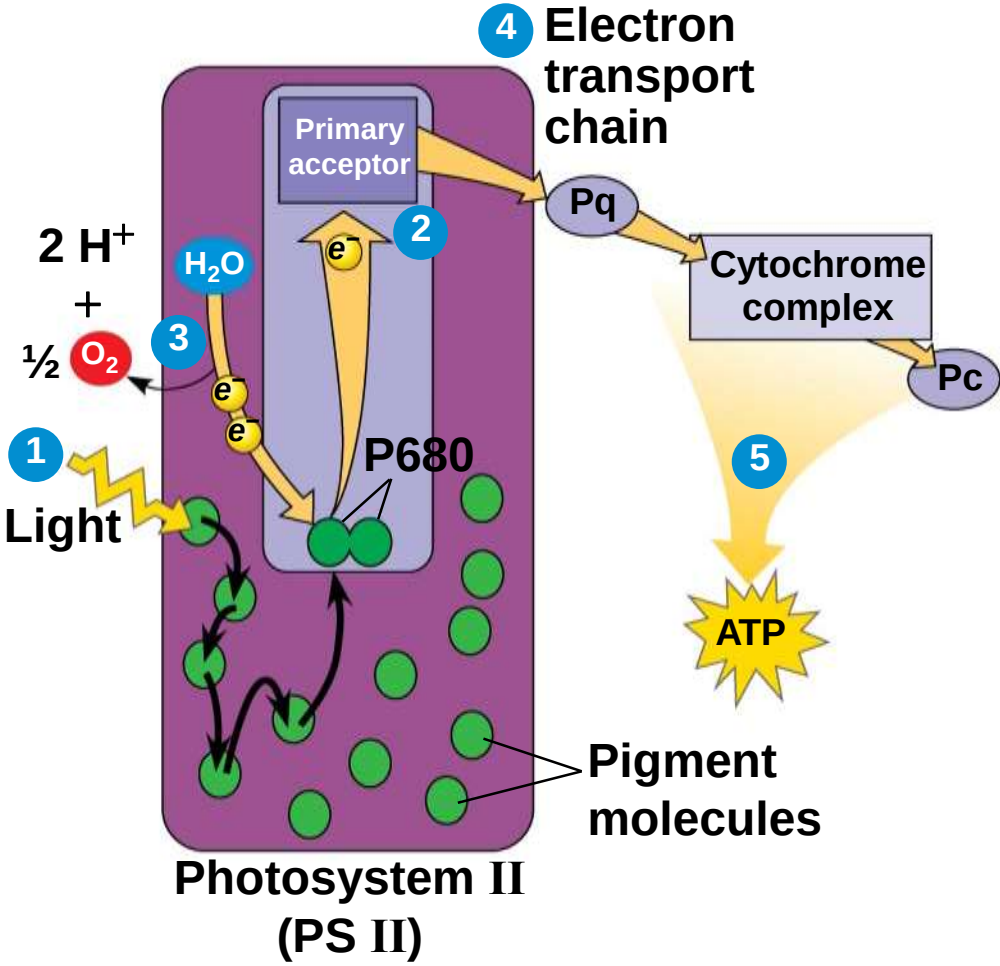
O_2 is released as a by-product of this reaction

Figure 10.14-2



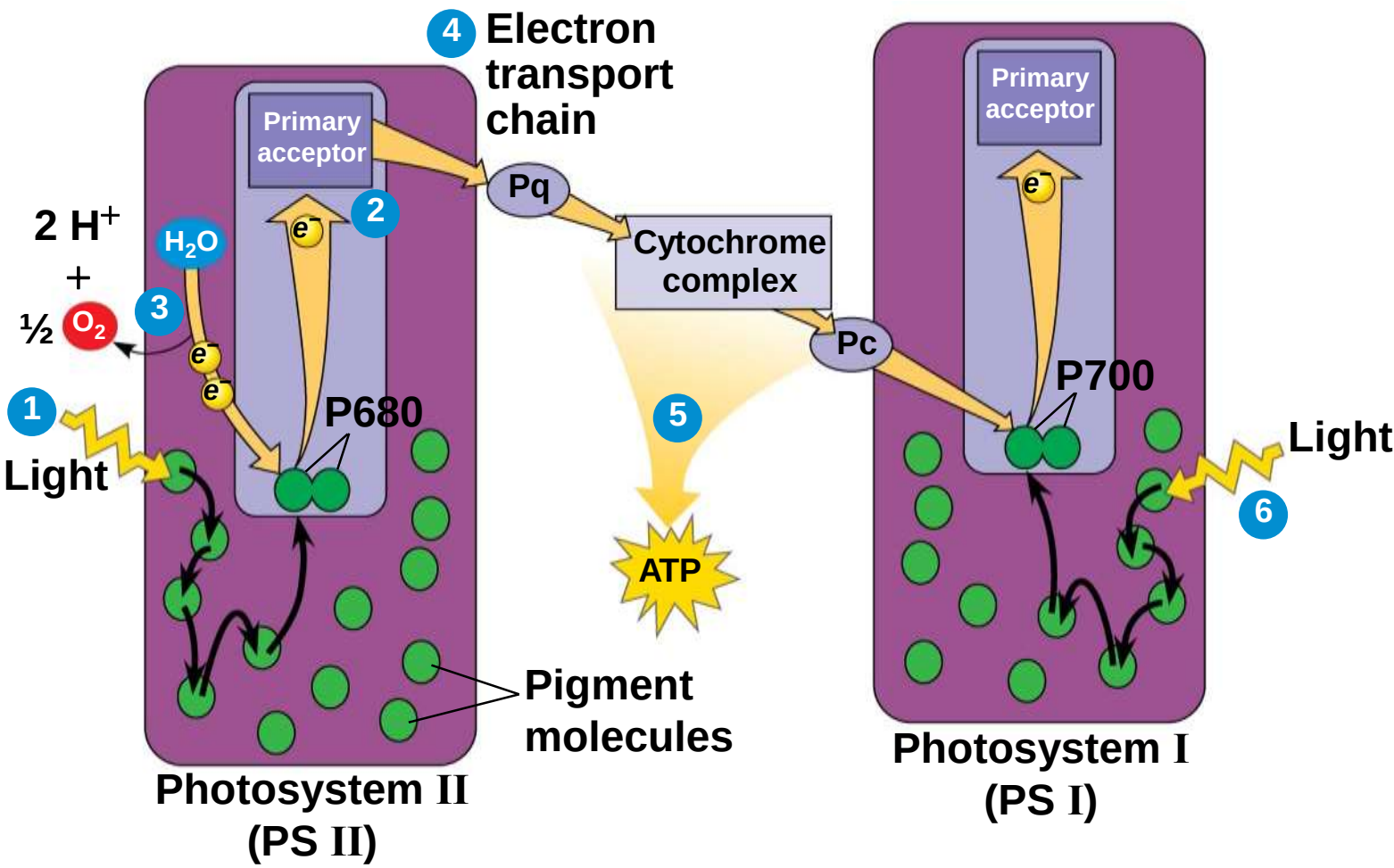
4. Each electron “falls” down an electron transport chain from the primary electron acceptor of PS II to PS I
5. Energy released by the fall drives the creation of a proton gradient across the thylakoid membrane
Diffusion of H^+ (protons) across the membrane drives ATP synthesis

Figure 10.14-3



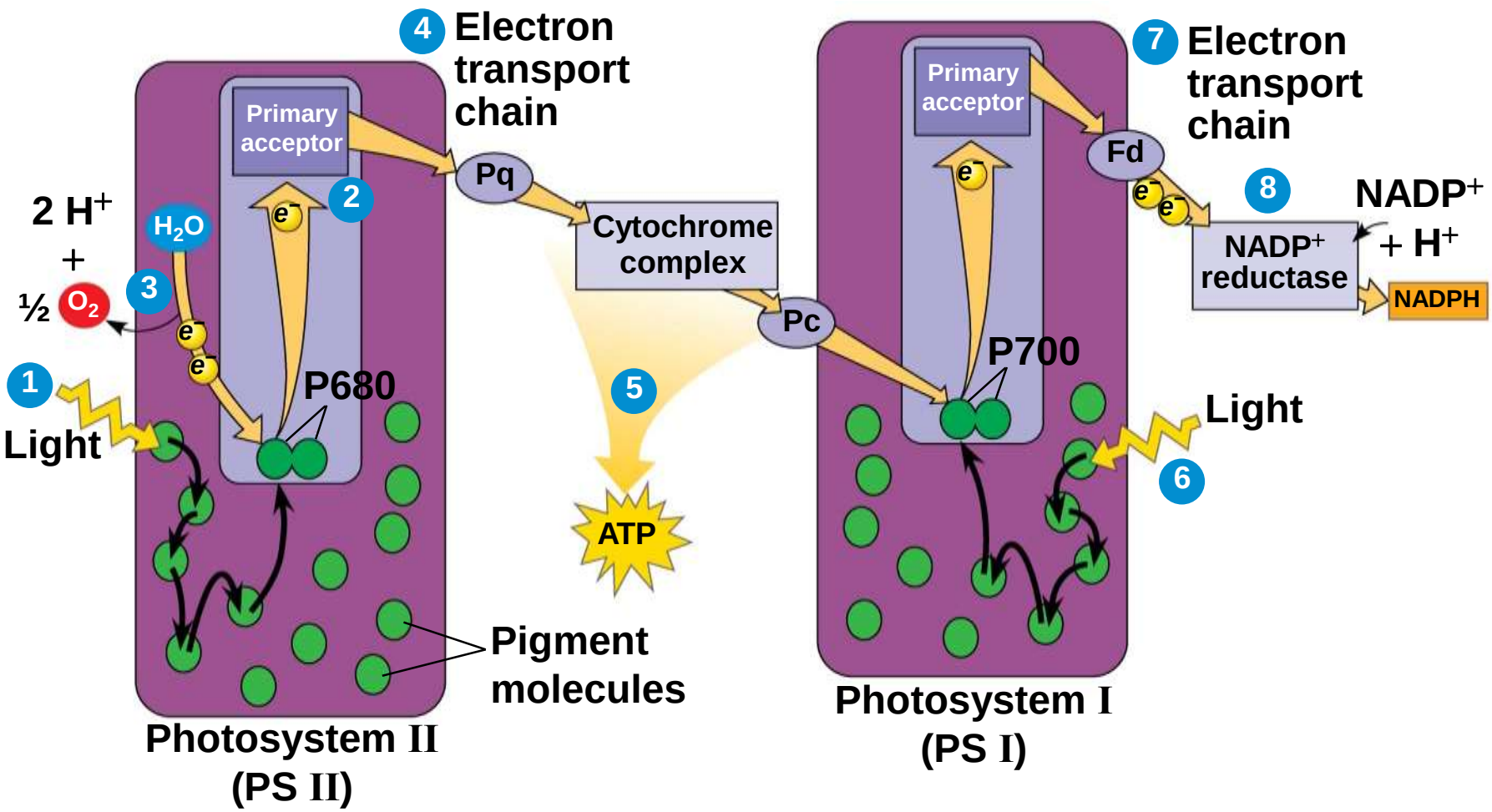
6. In PS I (like PS II), transferred light energy excites P700, which loses an electron to an electron acceptor
- a) $P700^+$ (P700 that is missing an electron) accepts an electron passed down from PS II via the electron transport chain

Figure 10.14-4



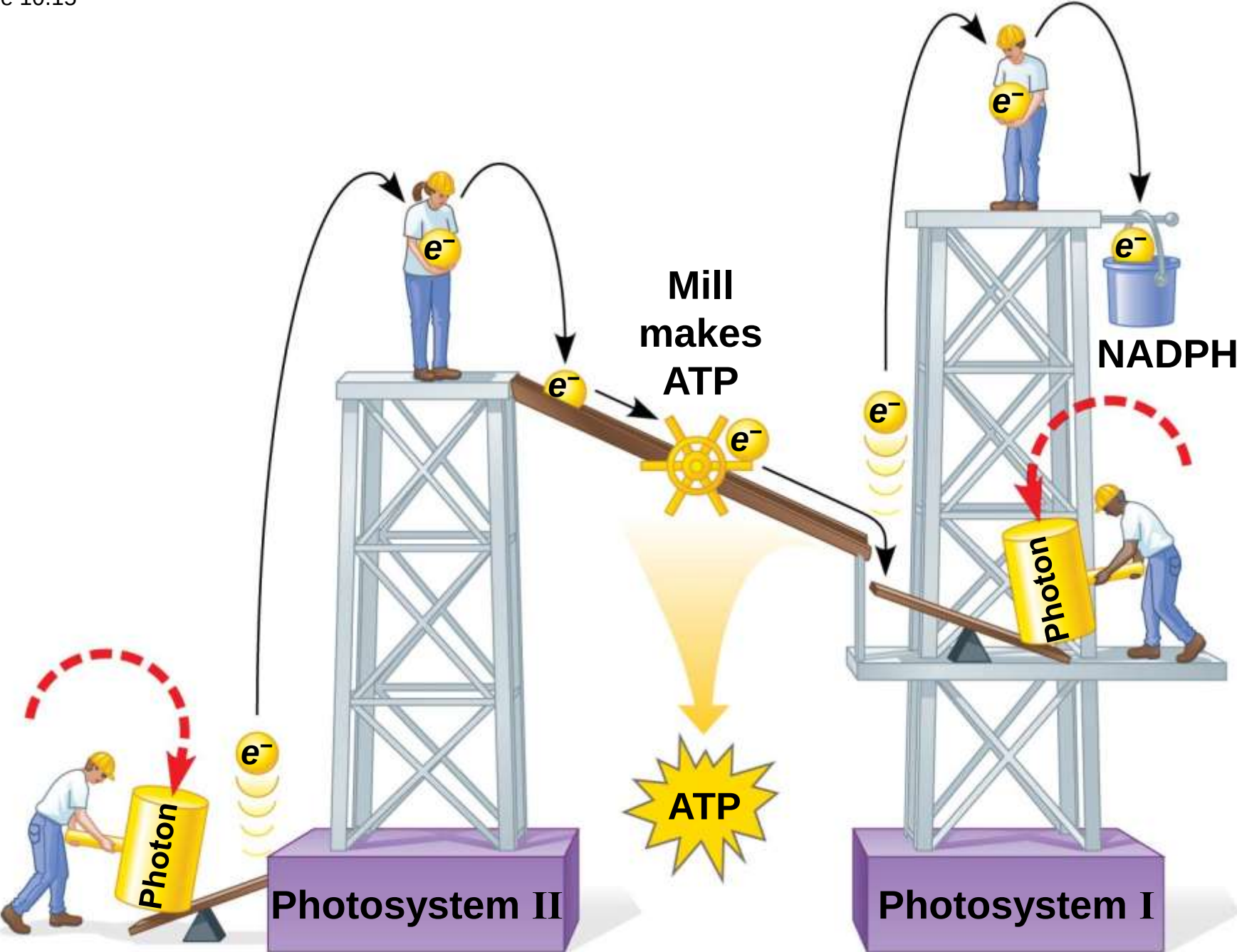
7. Each electron “falls” down an electron transport chain from the primary electron acceptor of PS I to the protein ferredoxin (Fd)
8. The electrons are then transferred to NADP^+ and reduce it to NADPH
 - a) The electrons of NADPH are available for the reactions of the Calvin cycle
 - b) This process also removes an H^+ from the stroma

Figure 10.14-5



a) The energy changes of electrons during linear flow through the light reactions can be shown in a mechanical analogy

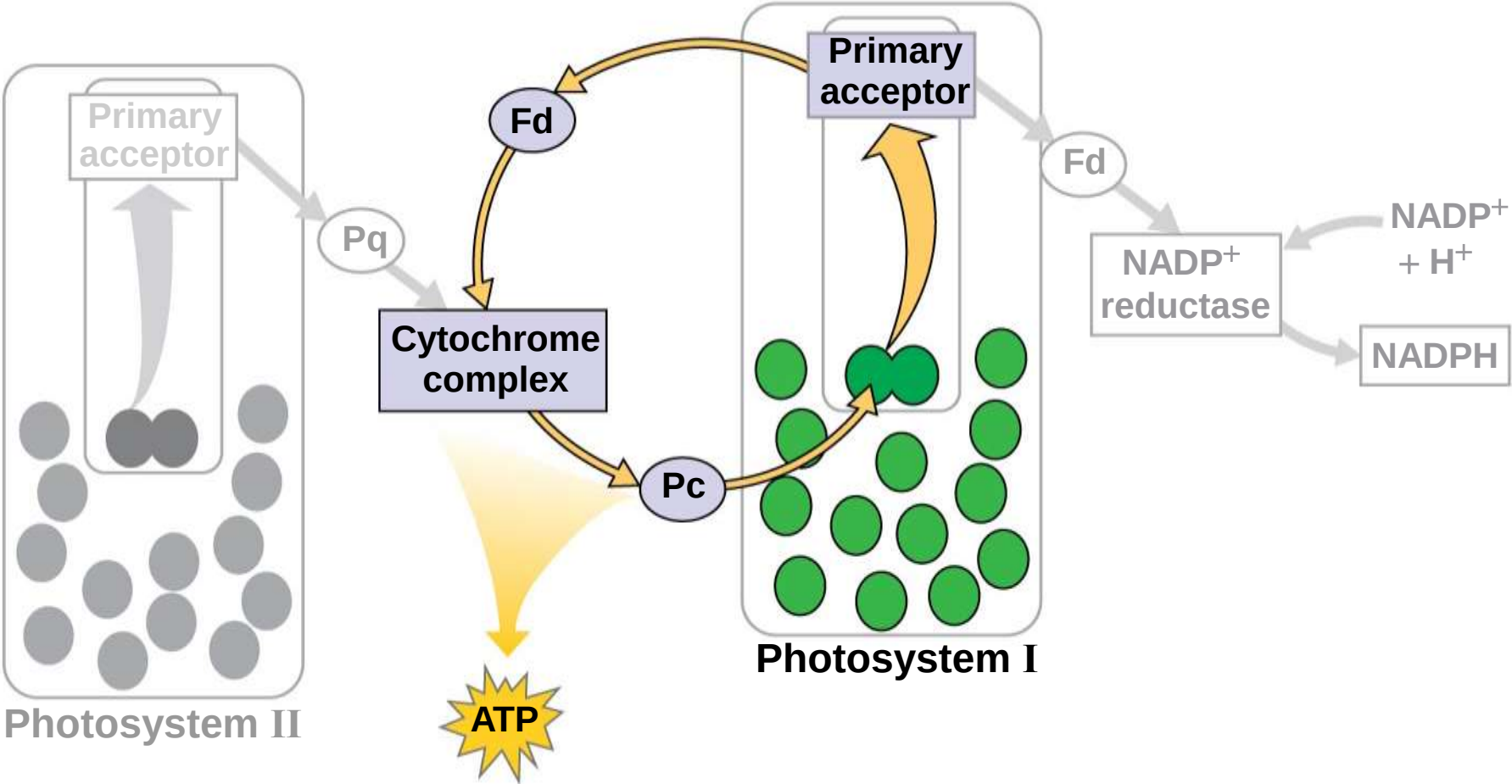
Figure 10.15



Cyclic Electron Flow

- a) In **cyclic electron flow**, electrons cycle back from Fd to the PS I reaction center
- b) Cyclic electron flow uses only photosystem I and produces ATP, but not NADPH
- c) No oxygen is released

Figure 10.16



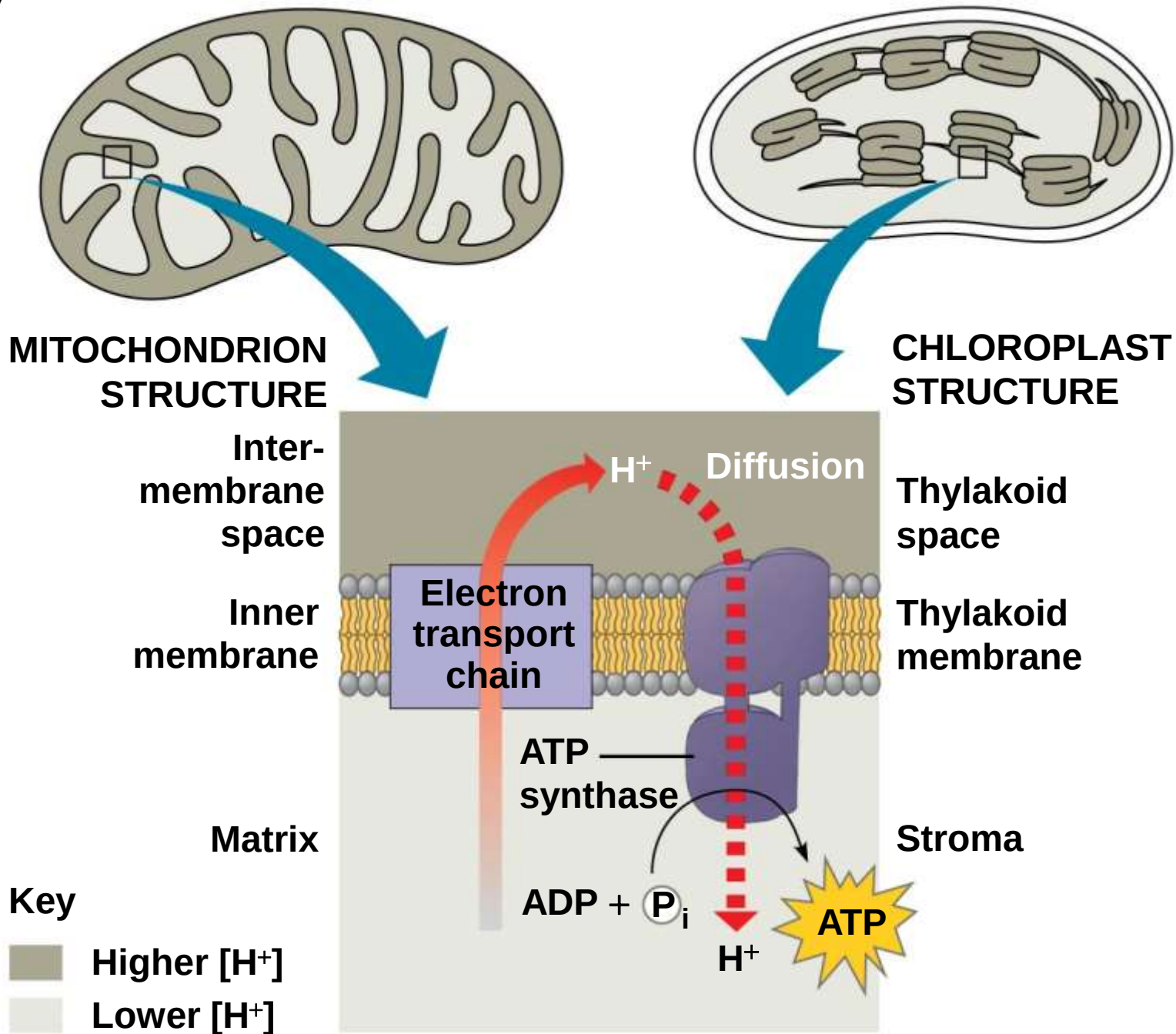
- a) Some organisms such as purple sulfur bacteria have PS I but not PS II
- b) Cyclic electron flow is thought to have evolved before linear electron flow
- c) Cyclic electron flow may protect cells from light-induced damage

A Comparison of Chemiosmosis in Chloroplasts and Mitochondria

- a) Chloroplasts and mitochondria generate ATP by chemiosmosis, but use different sources of energy
- b) Mitochondria transfer chemical energy from food to ATP; chloroplasts transform light energy into the chemical energy of ATP
- c) Spatial organization of chemiosmosis differs between chloroplasts and mitochondria but also shows similarities

- a) In mitochondria, protons are pumped to the intermembrane space and drive ATP synthesis as they diffuse back into the mitochondrial matrix
- b) In chloroplasts, protons are pumped into the thylakoid space and drive ATP synthesis as they diffuse back into the stroma

Figure 10.17



- a) ATP and NADPH are produced on the side facing the stroma, where the Calvin cycle takes place
- b) In summary, light reactions generate ATP and increase the potential energy of electrons by moving them from H₂O to NADPH

Figure 10.18

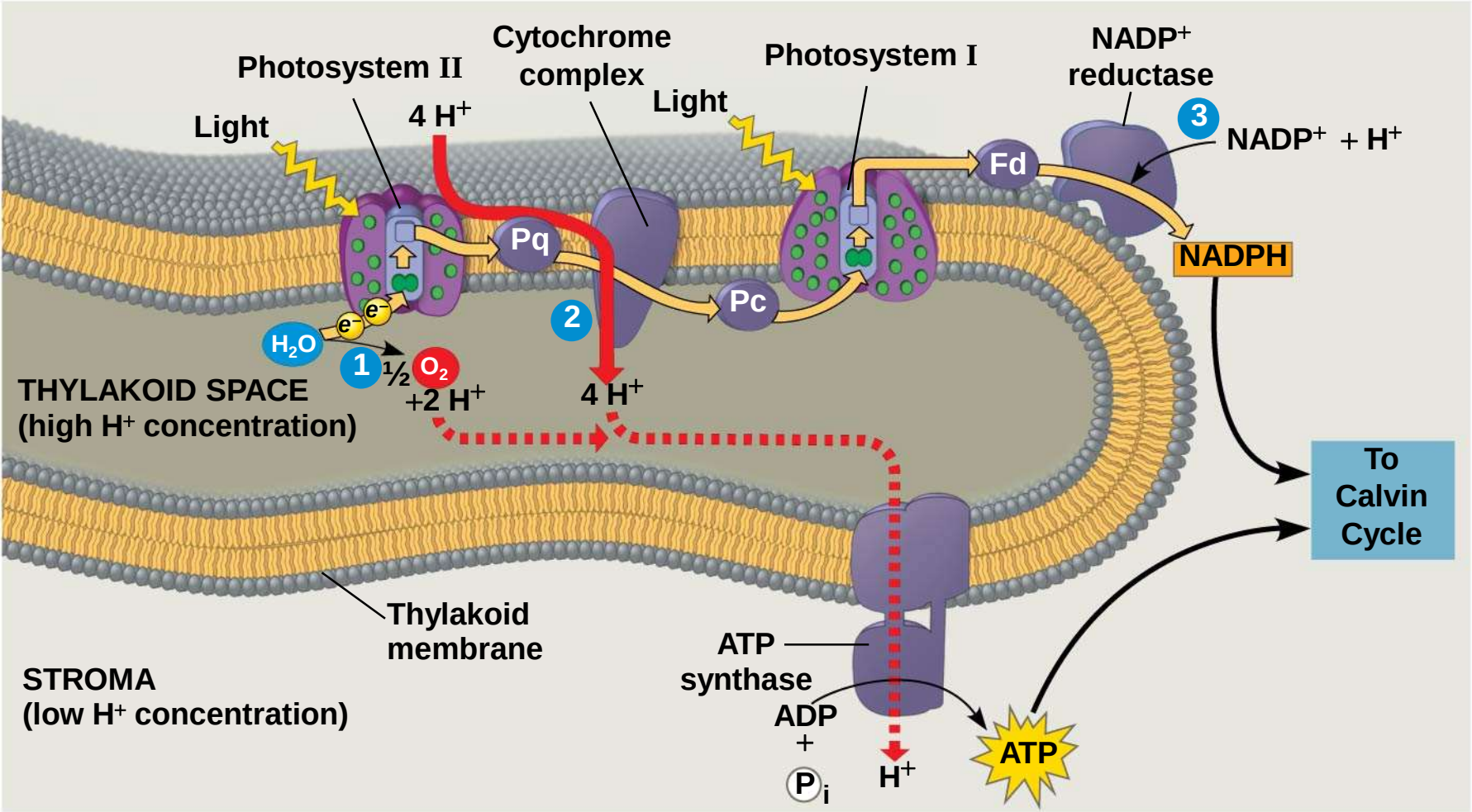


Figure 10.18a

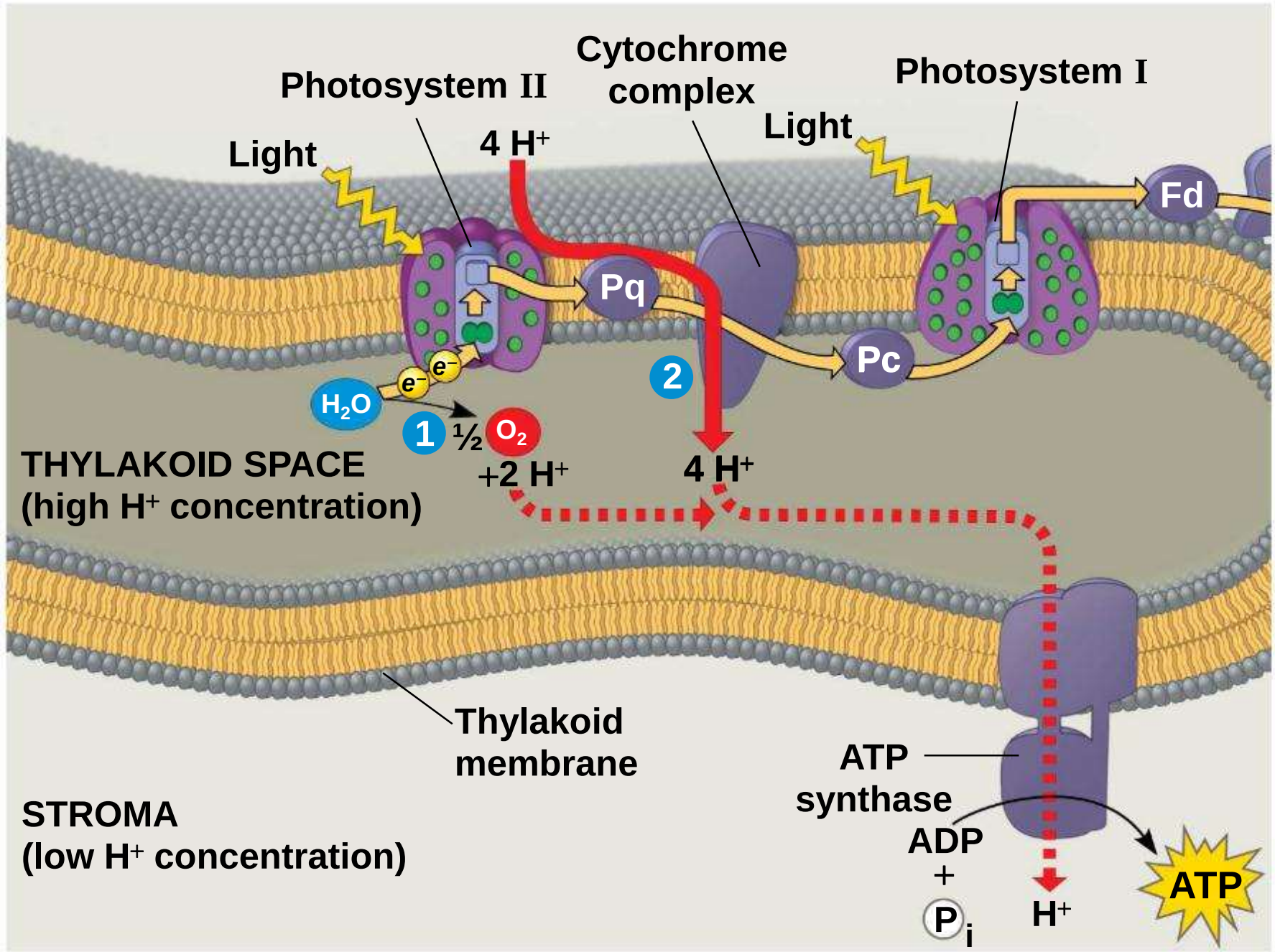
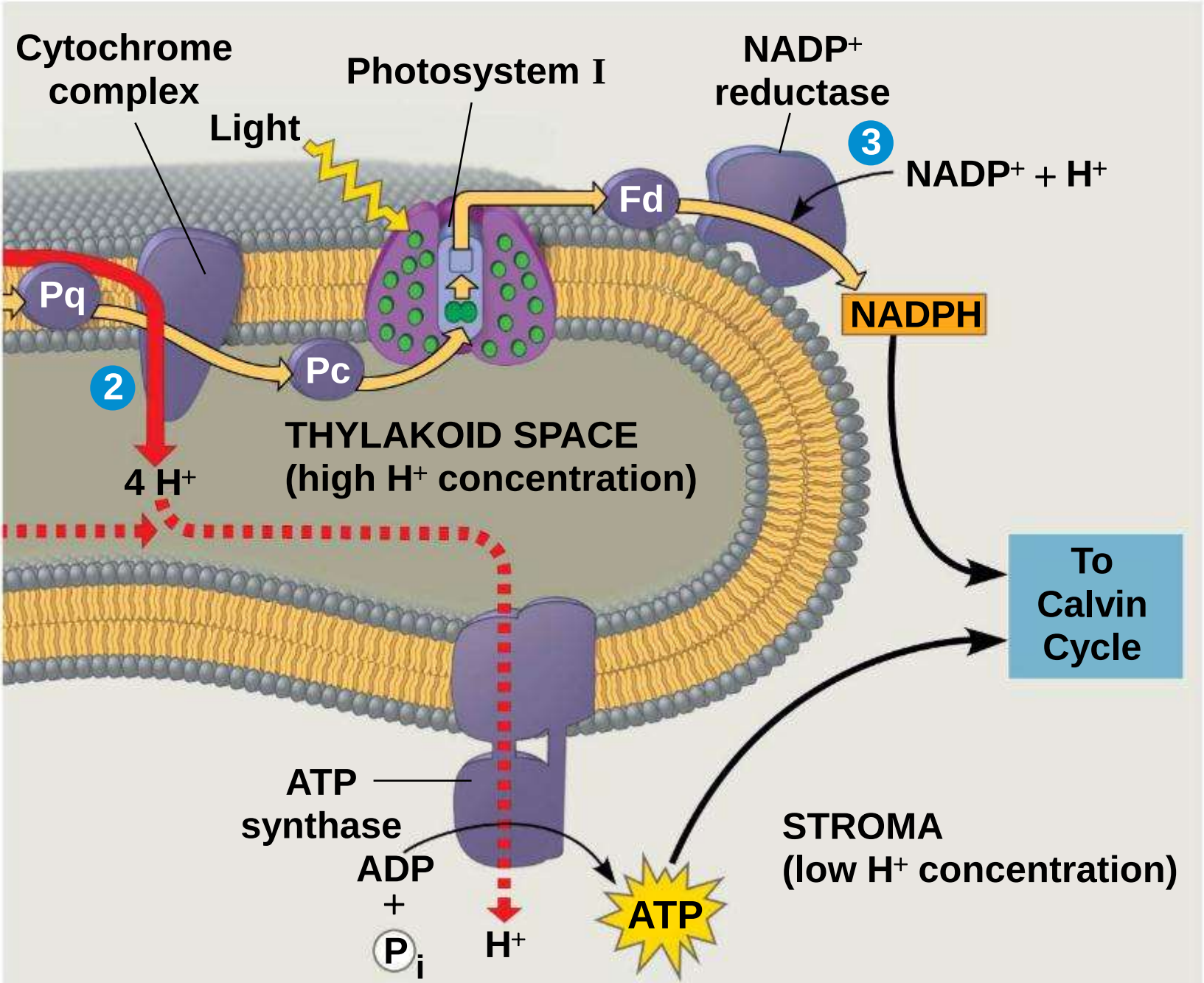


Figure 10.18b



Concept 10.3: The Calvin cycle uses the chemical energy of ATP and NADPH to reduce CO₂ to sugar

- a) The Calvin cycle, like the citric acid cycle, regenerates its starting material after molecules enter and leave the cycle
- b) The cycle builds sugar from smaller molecules by using ATP and the reducing power of electrons carried by NADPH

- a) Carbon enters the cycle as CO_2 and leaves as a sugar named **glyceraldehyde 3-phosphate (G3P)**
- b) For net synthesis of 1 G3P, the cycle must take place three times, fixing 3 molecules of CO_2
- c) The Calvin cycle has three phases
 1. **Carbon fixation** (catalyzed by **rubisco**)
 2. **Reduction**
 3. **Regeneration of the CO_2 acceptor (RuBP)**

Figure 10.UN03

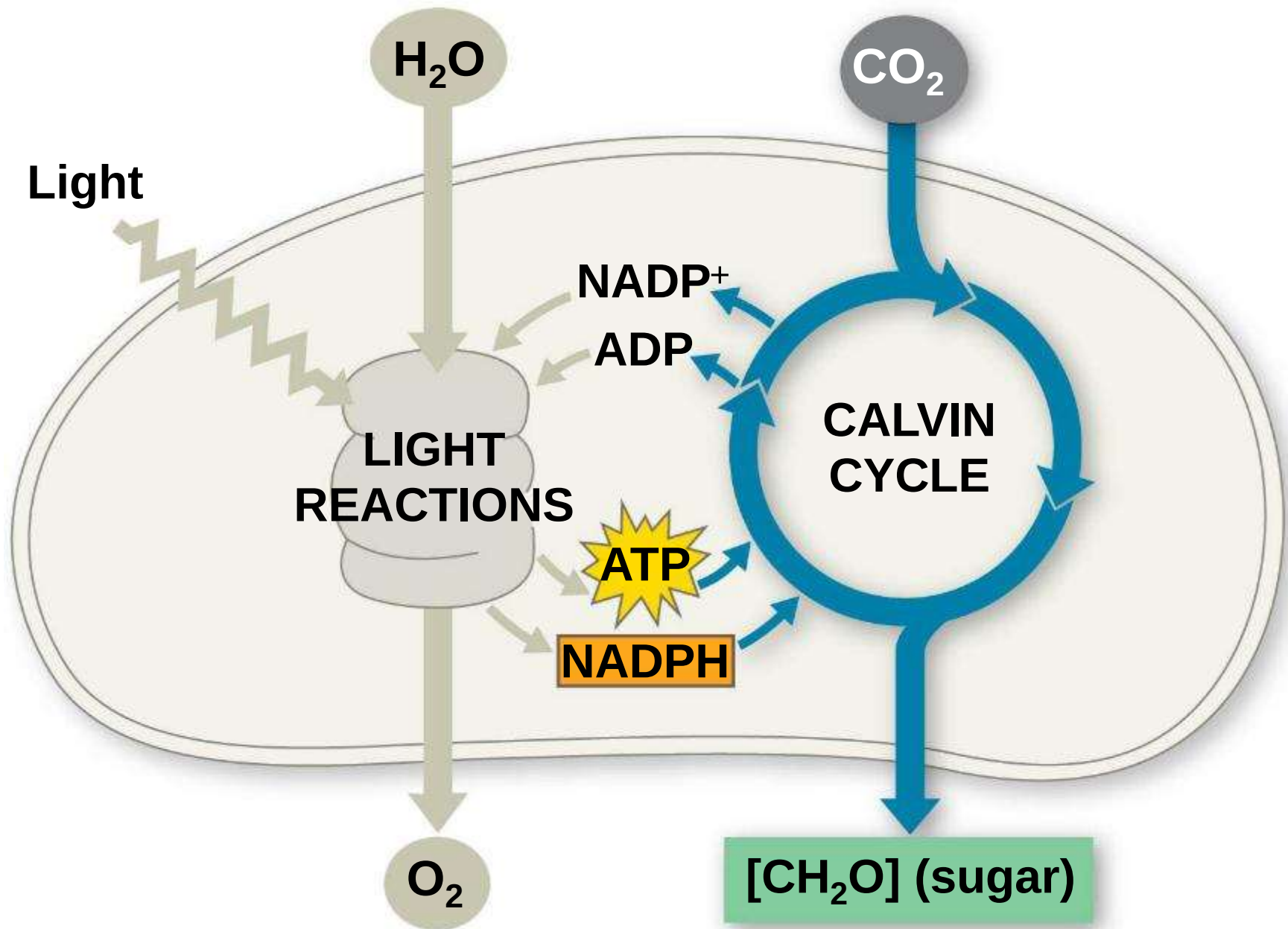


Figure 10.19-1

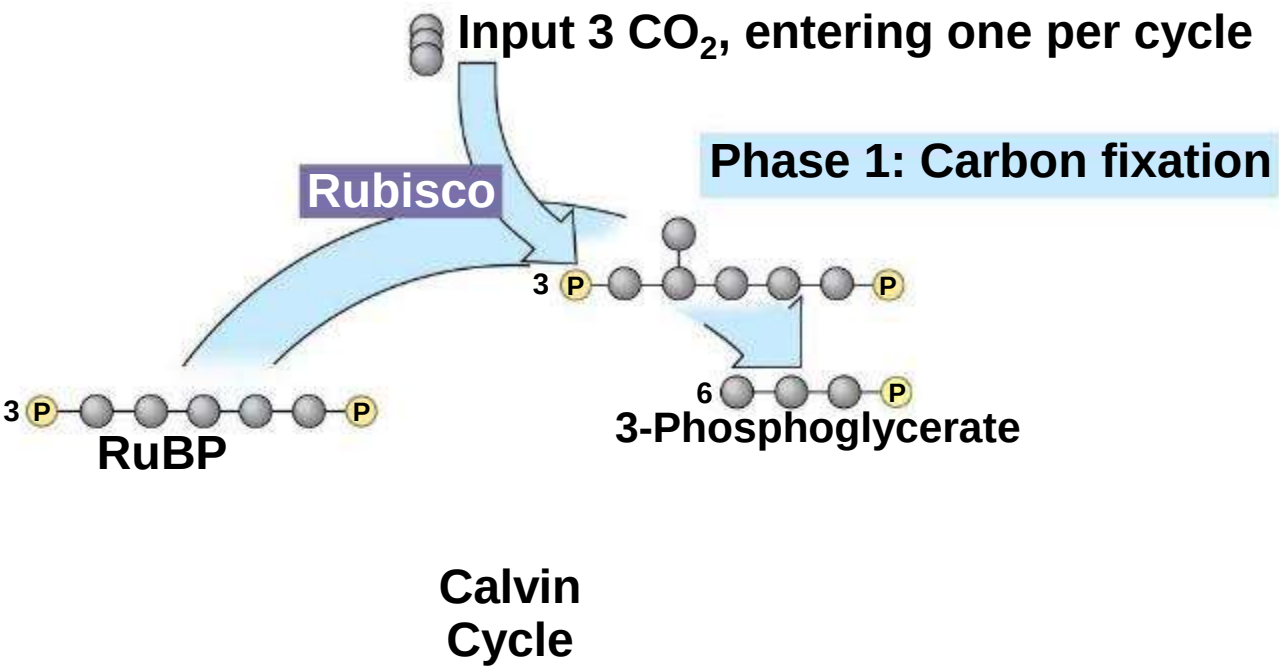


Figure 10.19-2

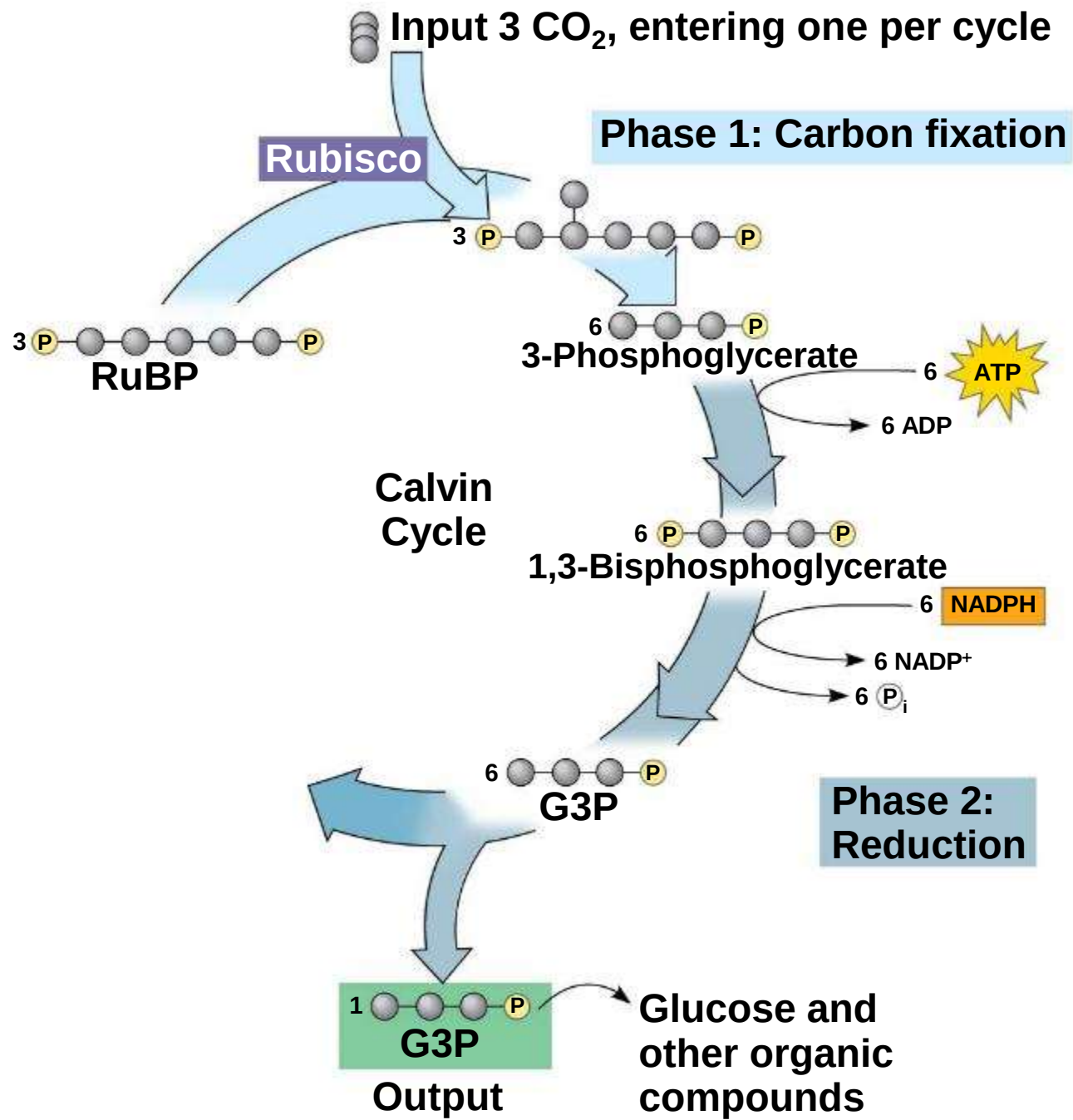
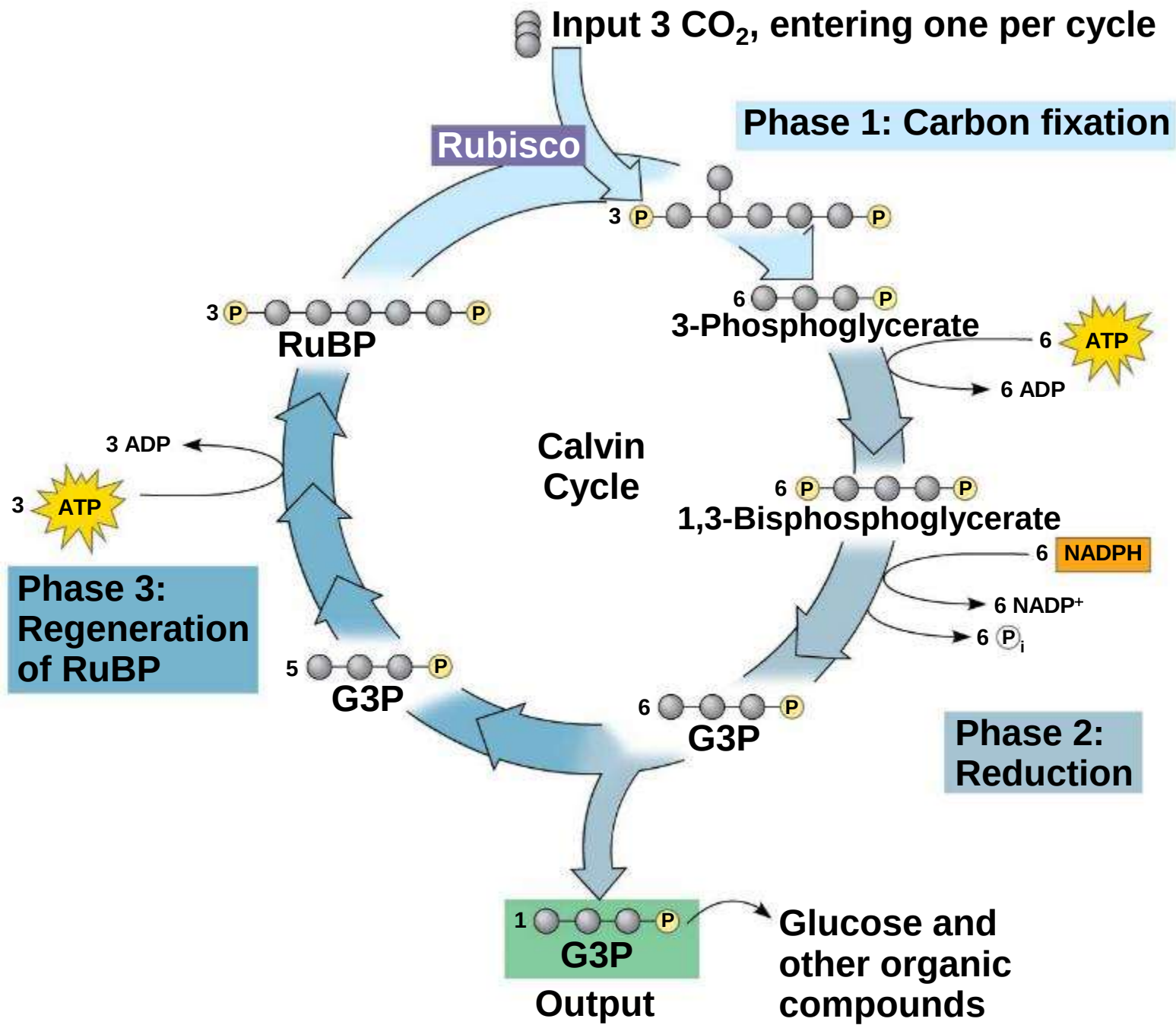


Figure 10.19-3



Concept 10.4: Alternative mechanisms of carbon fixation have evolved in hot, arid climates

- a) Dehydration is a problem for plants, sometimes requiring trade-offs with other metabolic processes, especially photosynthesis
- b) On hot, dry days, plants close stomata, which conserves H_2O but also limits photosynthesis
- c) The closing of stomata reduces access to CO_2 and causes O_2 to build up
- d) These conditions favor an apparently wasteful process called photorespiration

Photorespiration: An Evolutionary Relic?

- a) In most plants (**C₃ plants**), initial fixation of CO₂, via rubisco, forms a three-carbon compound (3-phosphoglycerate)
- b) In **photorespiration**, rubisco adds O₂ instead of CO₂ in the Calvin cycle, producing a two-carbon compound
- c) Photorespiration consumes O₂ and organic fuel and releases CO₂ without producing ATP or sugar

- a) Photorespiration may be an evolutionary relic because rubisco first evolved at a time when the atmosphere had far less O₂ and more CO₂
- b) Photorespiration limits damaging products of light reactions that build up in the absence of the Calvin cycle
- c) In many plants, photorespiration is a problem because on a hot, dry day it can drain as much as 50% of the carbon fixed by the Calvin cycle

C₄ Plants

- a) **C₄ plants** minimize the cost of photorespiration by incorporating CO₂ into four-carbon compounds e.g. sugarcane and corn
- b) There are two distinct types of cells in the leaves of C₄ plants:
 - a) **Bundle-sheath cells** are arranged in tightly packed sheaths around the veins of the leaf
 - b) Mesophyll cells are loosely packed between the bundle sheath and the leaf surface

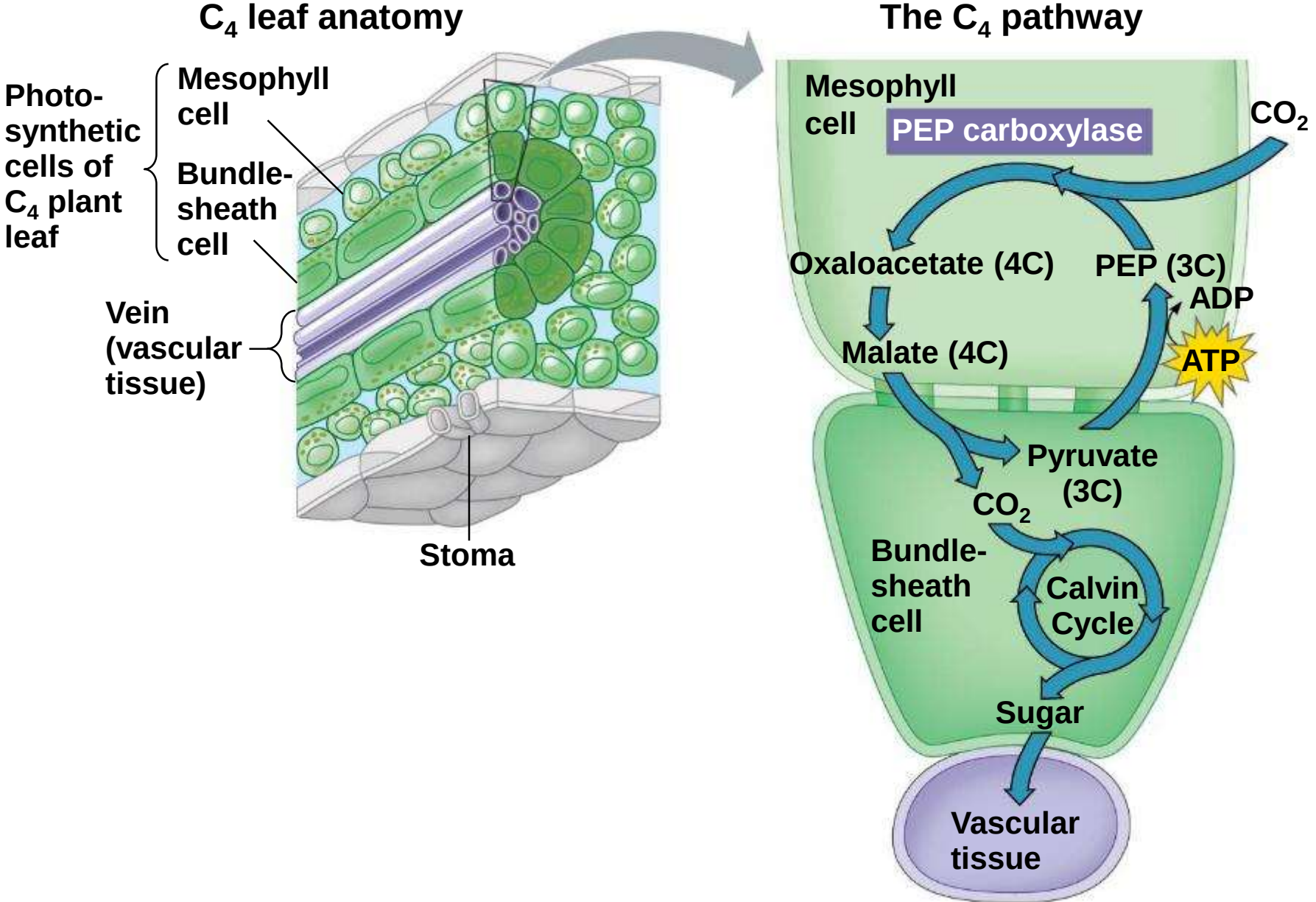
a) Sugar production in C_4 plants occurs in a three-step process:

1. The production of the four carbon precursors (oxaloacetate from PEP) is catalyzed by the enzyme **PEP carboxylase** in the mesophyll cells

- a) PEP carboxylase has a higher affinity for CO_2 than rubisco does; it can fix CO_2 even when CO_2 concentrations are low

2. These four-carbon compounds are exported to bundle-sheath cells
3. Within the bundle-sheath cells, they release CO₂ that is then used in the Calvin cycle

Figure 10.20



- a) Since the Industrial Revolution in the 1800s, CO₂ levels have risen greatly
- b) Increasing levels of CO₂ may affect C₃ and C₄ plants differently, perhaps changing the relative abundance of these species
- c) The effects of such changes are unpredictable and a cause for concern

CAM Plants

- a) Some plants, including succulents (water-storing), use **crassulacean acid metabolism (CAM)** to fix carbon
- b) **CAM plants** open their stomata at night, incorporating CO_2 into organic acids
- c) Stomata close during the day, and CO_2 is released from organic acids and used in the Calvin cycle

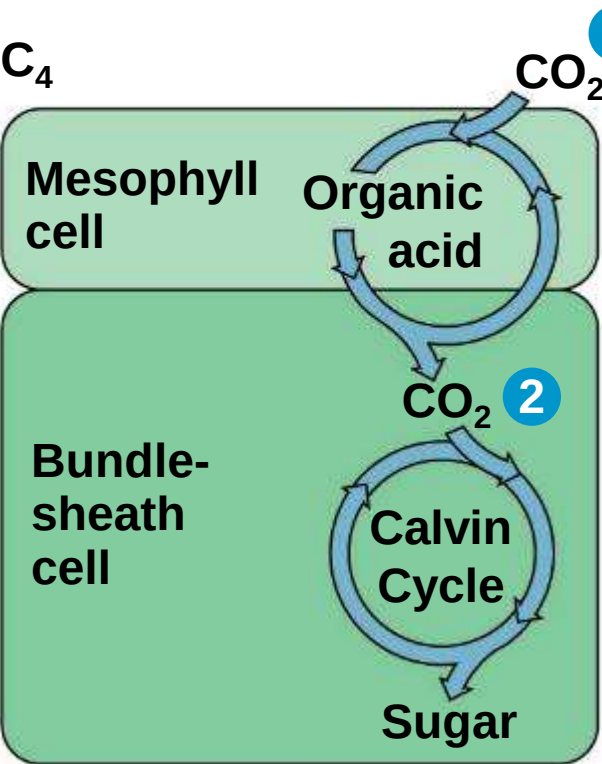
Figure 10.21



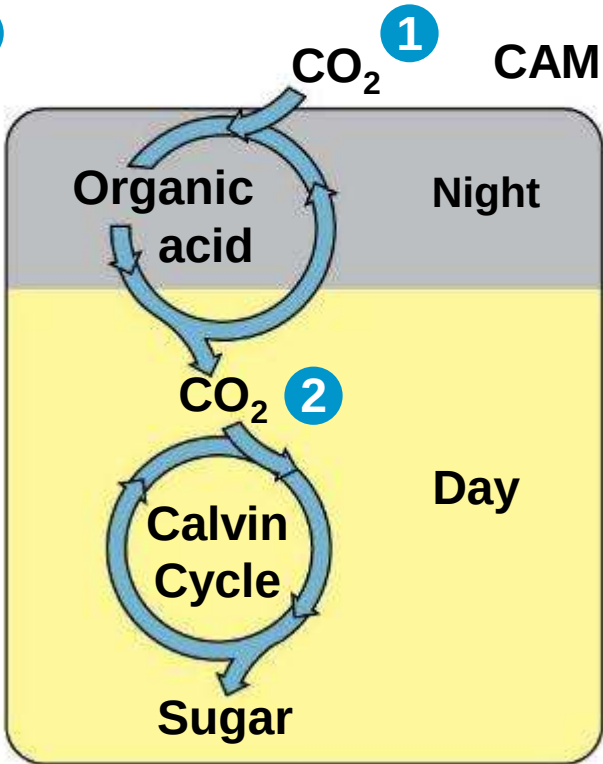
Sugarcane



Pineapple



(a) Spatial separation of steps

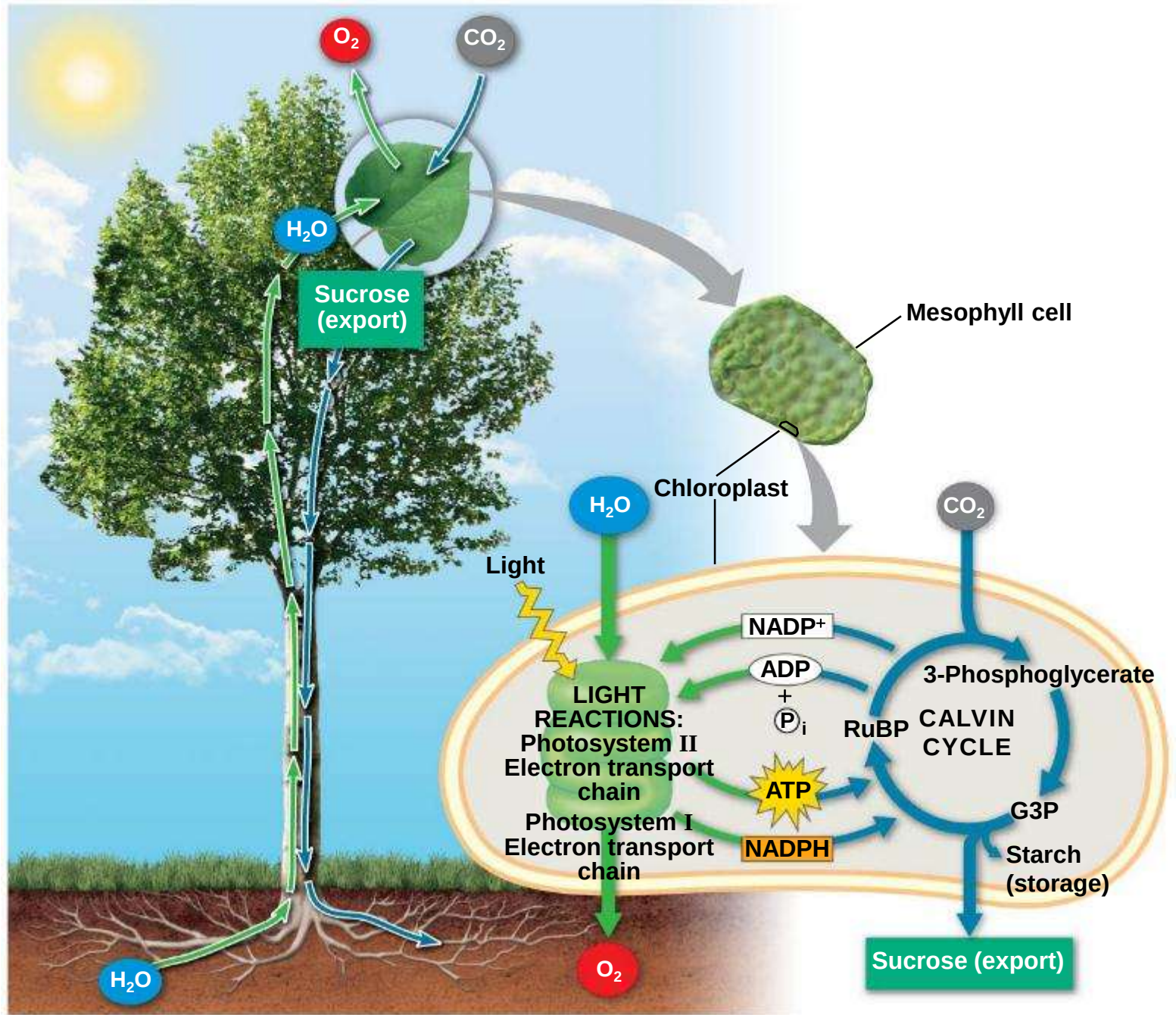


(b) Temporal separation of steps

The Importance of Photosynthesis: *A Review*

- a) The energy entering chloroplasts as sunlight gets stored as chemical energy in organic compounds
- b) Sugar made in the chloroplasts supplies chemical energy and carbon skeletons to synthesize the organic molecules of cells
- c) Plants store excess sugar as starch in structures such as roots, tubers, seeds, and fruits
- d) In addition to food production, photosynthesis produces the O₂ in our atmosphere

Figure 10.22a



LIGHT REACTIONS

- **Are carried out by molecules in the thylakoid membranes**
- **Convert light energy to the chemical energy of ATP and NADPH**
- **Split H_2O and release O_2 to the atmosphere**

CALVIN CYCLE REACTIONS

- **Take place in the stroma**
- **Use ATP and NADPH to convert CO_2 to the sugar G3P**
- **Return ADP, inorganic phosphate, and NADP^+ to the light reactions**

Figure 10.23

MAKE CONNECTIONS
The Working Cell

Flow of Genetic Information
in the Cell:
DNA → RNA → Protein
(Chapters 5–7)

Movement Across
Cell Membranes
(Chapter 7)

Energy Transformations in the Cell:
Photosynthesis and Cellular
Respiration (Chapters 8–10)

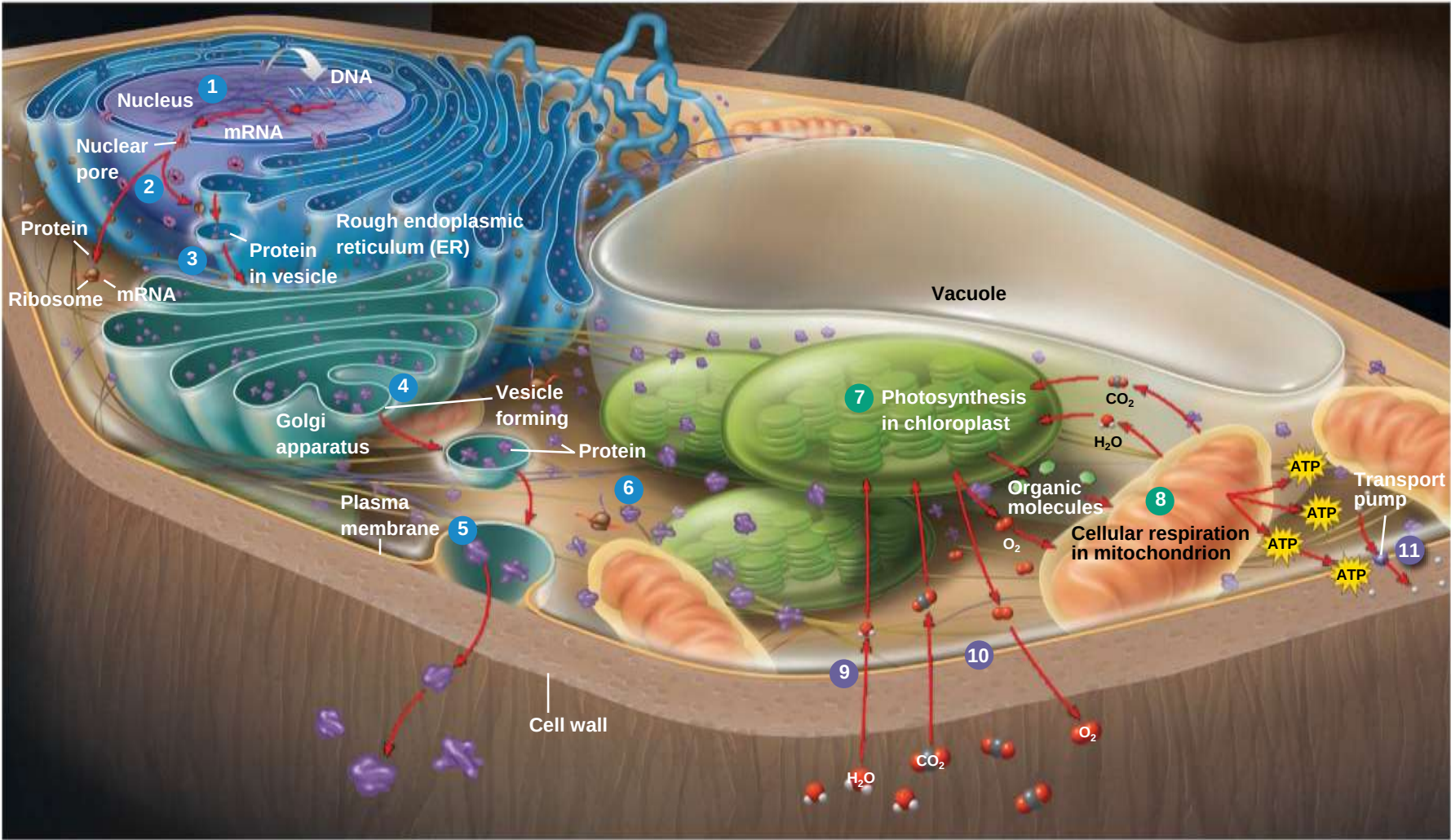


Figure 10.23a

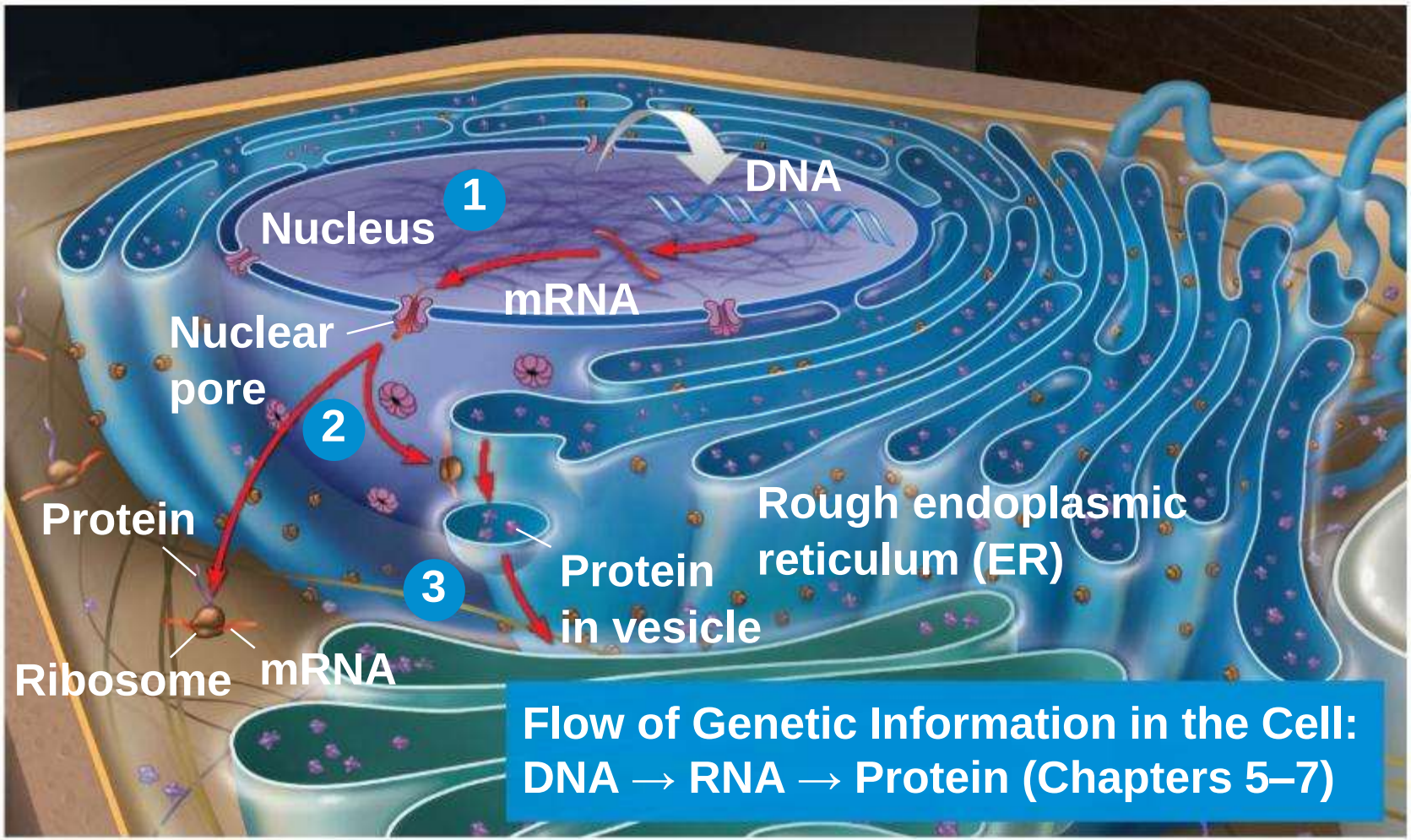


Figure 10.23b

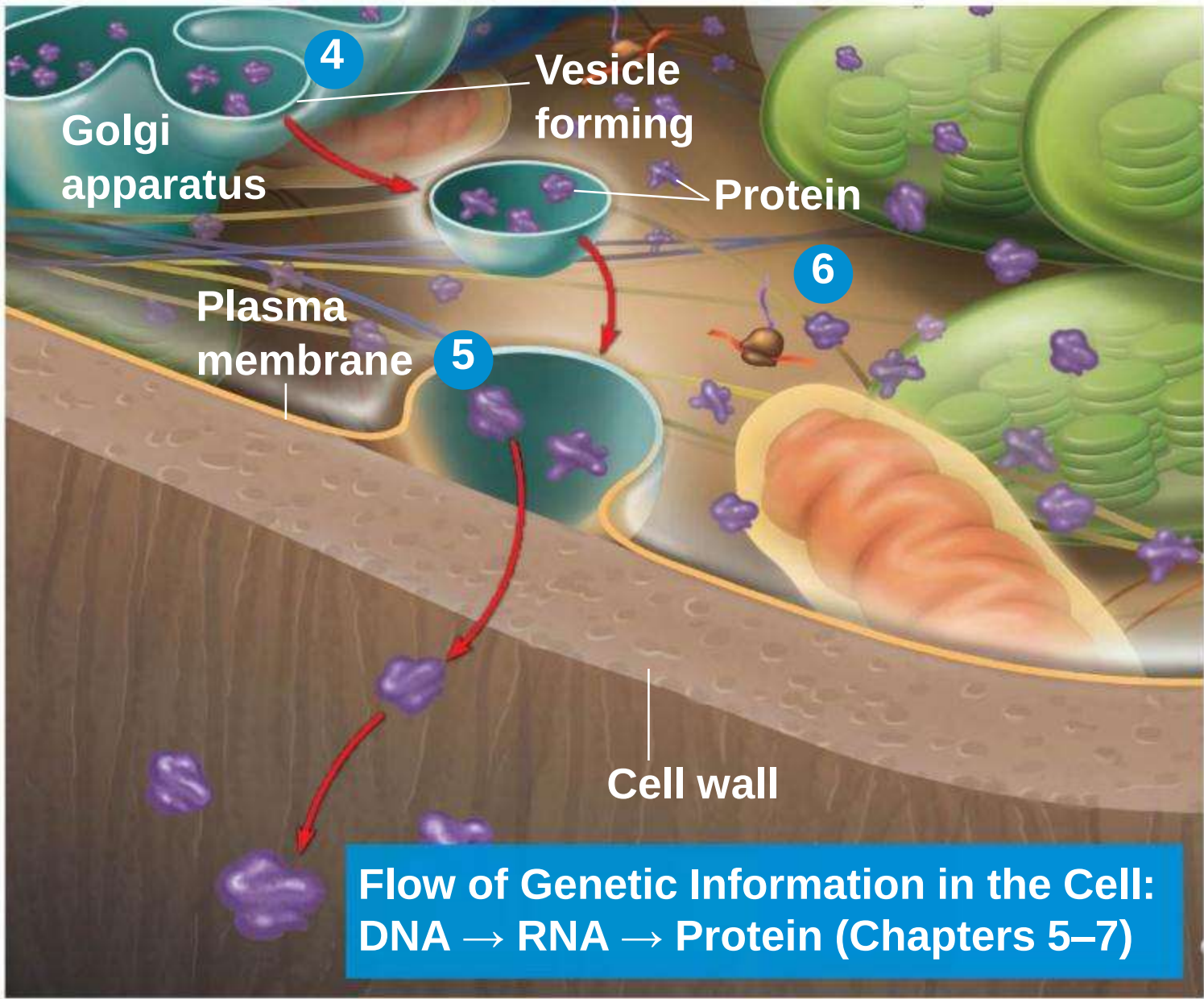


Figure 10.23c

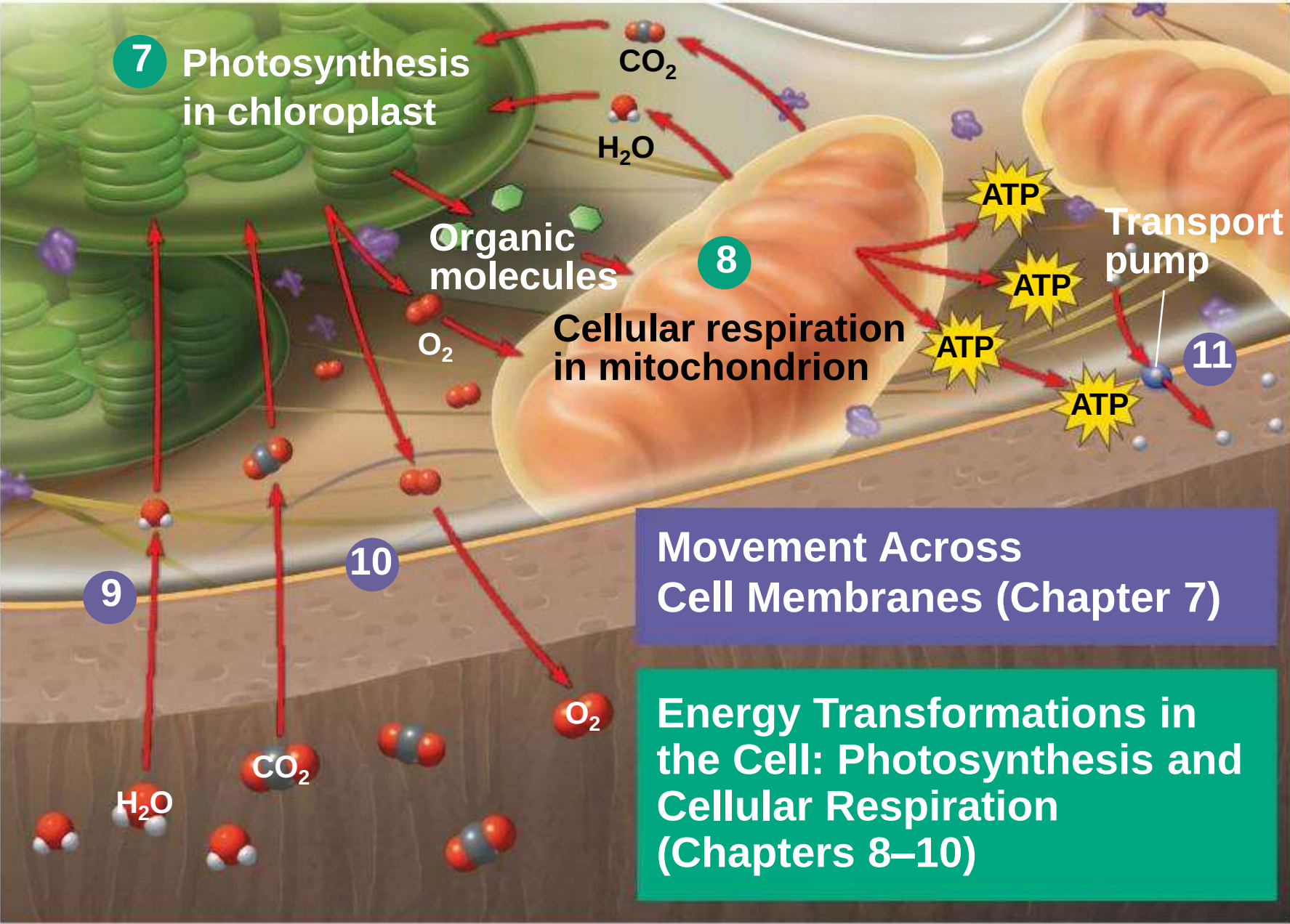


Figure 10.UN04a

	350 ppm CO ₂	600 ppm CO ₂	1,000 ppm CO ₂
Average dry mass of one corn plant (g)	91	89	80
Average dry mass of one velvetleaf plant (g)	35	48	54

Figure 10.UN05

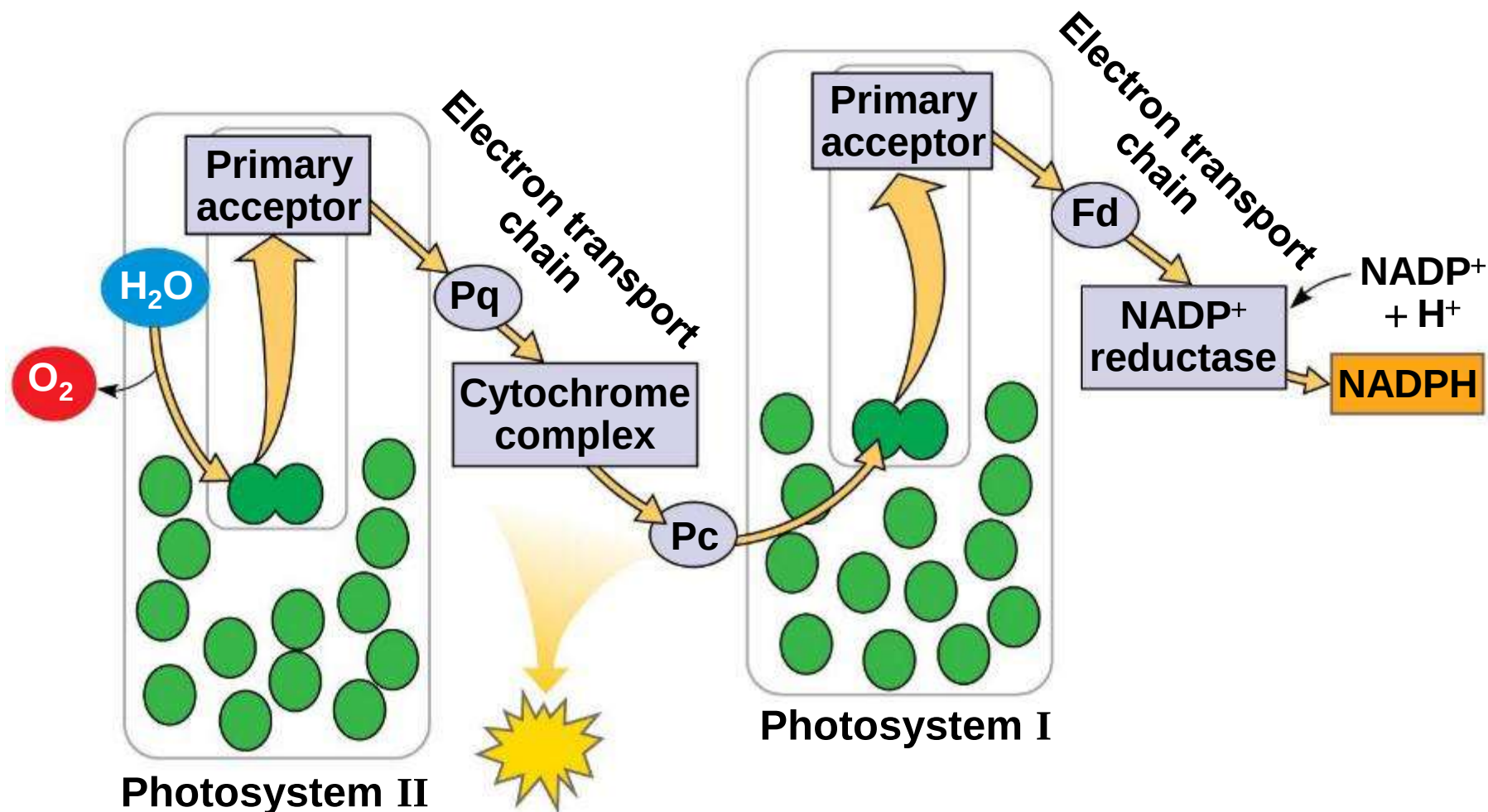


Figure 10.UN06

